HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

# GRADUATION THESIS

## Multi-Agent Medical Assistant with Vision-Language, Image Segmentation, Knowledge Retrieval and Persistent Memory

**HOANG QUOC HUNG**
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**Program: Global ICT**

**Supervisor:** PhD. Dinh Viet Sang
**Department:** Department of Computer Science
**School:** School of Information and Communication Technology

**HANOI, 06/2026**

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# ABSTRACT

This thesis proposes a Multi-Agent Medical Assistant for medical consultation across text and image inputs. The starting point is a practical problem: medical questions rarely arrive in a clean textbook form. A user may describe symptoms incompletely, ask follow-up questions across several sessions, or upload an image that requires both visual interpretation and cautious medical explanation.

Instead of relying on a single generative model, the system routes user requests to specialized agents, combines structured and unstructured medical evidence, checks whether the available clinical context is sufficient, and preserves useful patient context without treating it as a verified diagnosis. In the text pathway, the system coordinates Retrieval-Augmented Generation, Knowledge Graph reasoning, multi-turn clarification, MedGemma CoT-based information sufficiency assessment and answer synthesis, and Web Search fallback. In the vision pathway, it combines three-class colonoscopy polyp segmentation, mask post-processing, and Polyp Visual Question Answering. The segmentation mask is used as a spatial grounding prior, while the original endoscopic image remains the source of visual evidence.

Experimental results show that the proposed design achieves several system-level goals. Knowledge Graph evidence improves relation-oriented text reasoning and multi-turn awareness; post-processing improves segmentation consistency for downstream use; LoRA fine-tuning substantially improves Polyp VQA performance; segmentation grounding improves judge-based VQA quality on the mask-matched subset; and patient-scoped memory improves routing accuracy, repeated-question avoidance, and answer quality while introducing latency and relevance-filtering trade-offs. Overall, the thesis demonstrates that a modular, grounded, memory-aware architecture can improve observability, evidence control, multimodal reasoning, and safety boundaries in a medical assistant intended for decision support rather than definitive clinical diagnosis.

**Keywords:** Multi-agent medical assistant, retrieval-augmented generation, knowledge graph, medical visual question answering, visual grounding, long-term memory, human-in-the-loop validation.

Student

*(Signature and full name)*

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# CHAPTER 1. INTRODUCTION

## 1.1 Motivation

Healthcare information seeking is increasingly mediated by digital systems, yet most users still face a fundamental gap between everyday symptom descriptions and reliable medical interpretation. A patient may say that they have stomach pain, dizziness, or chest discomfort, but leave out details that strongly affect the next safe step: onset time, duration, severity, medication use, pregnancy status, allergies, chronic disease history, or warning signs. At the same time, medical information available on the internet is abundant but uneven in quality, difficult to verify, and often not adapted to the user's clinical context.

Recent progress in large language models (LLMs), retrieval-augmented generation (RAG), knowledge graphs, and vision-language models (VLMs) makes it possible to design medical assistants that can reason over both text and images. However, a direct application of a single generative model is not sufficient for a safety-critical domain. Medical consultation requires source grounding, uncertainty awareness, follow-up questioning, multimodal interpretation, and mechanisms for human validation.

This thesis develops a **Multi-Agent Medical Assistant** that combines text-based medical reasoning, medical image analysis, long-term memory, and human-in-the-loop validation. The system is designed as a coordinated set of specialized agents rather than a single monolithic model. In practice, this means the assistant may ask before it answers, retrieve evidence before it explains, or route an image to a specialized visual workflow instead of treating every request as ordinary chat. In the text pathway, structured Knowledge Graph evidence and unstructured RAG evidence are coordinated before MedGemma CoT performs information-sufficiency assessment and final answer synthesis. In the vision pathway, polyp segmentation provides spatial grounding for VQA rather than replacing direct interpretation of the original endoscopic image. The memory layer supplies patient-scoped contextual continuity while remaining separate from verified clinical diagnosis.

## 1.2 Problem Statement

The problem addressed in this thesis is how to design and evaluate a medical assistant that remains grounded, modular, and cautious across heterogeneous user inputs. The system must not only generate answers, but also decide when evidence is sufficient, when follow-up information is required, and how text, image, memory, and validation signals should be coordinated. Concretely, the thesis investigates whether a multi-agent architecture can:

- produce medical answers grounded in structured and unstructured evidence;
- detect incomplete clinical context and collect missing information through multi-turn clarification;
- coordinate RAG, Knowledge Graph reasoning, MedGemma CoT sufficiency

assessment and answer synthesis, and Web Search fallback for text-based consultation;
- connect general medical image analysis with a specialized colonoscopy polyp workflow;
- use polyp segmentation masks as spatial grounding for VQA while preserving the original image as the visual evidence source;
- maintain patient-scoped context across sessions without treating memory as verified diagnosis;
- support human-in-the-loop validation for review and safety feedback.

The focus is not on building a web application interface, but on the AI architecture, reasoning workflow, multimodal coordination, memory integration, and experimental analysis.

## 1.3 Technical Challenges

Several challenges motivate the proposed architecture.

**Incomplete clinical context.** Users often provide insufficient symptom descriptions. A reliable medical assistant should recognize when information is missing and ask follow-up questions before generating a recommendation.

**Hallucination and source reliability.** LLMs can generate fluent but unsupported medical content. Grounding responses in RAG documents, Knowledge Graph facts, and trusted external sources reduces this risk. However, retrieved evidence alone is not sufficient; the system must also decide whether the user's clinical context is complete enough for a safe answer.

**Structured and unstructured medical knowledge.** Medical reasoning often requires both detailed textual explanation and structured relations between symptoms, diseases, drugs, and risk factors. RAG and Knowledge Graphs provide complementary capabilities.

**Medical image understanding.** Medical image analysis requires models that can identify visual patterns and communicate findings carefully. In colonoscopy images, polyp localization and neoplasm characterization are important subtasks.

**Visual grounding for VQA.** A VLM answering questions from an image may attend to irrelevant regions. Segmentation masks can guide the VQA model toward clinically relevant regions by providing a spatial prior, but the final answer must still be derived from visual evidence in the original image. Incorrect masks may also introduce noise or misdirect the VQA model.

**Long-term patient context.** Medical interaction is rarely isolated. Information such as allergies, medication use, chronic diseases, or repeated symptoms may remain relevant across sessions. A memory layer must preserve useful facts without treating them as official diagnoses.

**Safety and validation.** Medical AI systems require feedback mechanisms and explicit boundaries. Human-in-the-loop validation helps collect feedback and identify responses that may require review.

## 1.4 Objectives

The main objective of this thesis is to design, implement, and evaluate an AI-centered multi-agent medical assistant that can support preliminary medical information seeking in a grounded, modular, and safety-aware manner. Instead of relying on a single generative model, the system aims to coordinate multiple specialized agents so that each type of input is handled by the most appropriate reasoning pathway. The assistant is expected to retrieve evidence before generating medical explanations, ask for additional information when the user's description is incomplete, incorporate patient-scoped context when relevant, and provide mechanisms for user validation and feedback.

More specifically, the thesis focuses on four core capabilities:

- **Text Module:** to build a consultation-oriented text reasoning workflow that combines the Conversation Agent, multi-turn clarification, query expansion, RAG, Knowledge Graph reasoning, MedGemma CoT-based information sufficiency assessment and answer synthesis, and Web Search fallback. This objective emphasizes reducing premature answers when the clinical context is incomplete and improving response grounding through both structured and unstructured evidence.
- **Vision Module:** to support medical image understanding through general image analysis, colonoscopy polyp segmentation, and grounded Polyp VQA. The goal is not only to answer visual questions, but also to investigate whether segmentation masks can provide useful spatial grounding for VQA while preserving the original image as the primary visual evidence source.
- **Memory Layer:** to integrate patient-scoped long-term context using a Mem0-style memory mechanism. This layer is designed to preserve useful information such as allergies, chronic conditions, medication history, or repeated symptoms across sessions, while keeping stored patient-reported facts separate from verified clinical diagnoses.
- **Human Validation:** to incorporate user validation and feedback logging as a human-in-the-loop safety mechanism. This objective supports review of generated responses, collection of correction signals, and analysis of cases where the assistant may need to be more cautious or request professional medical attention.

In addition to system design, the thesis also aims to evaluate the proposed architecture at both component and workflow levels. The evaluation covers text-based medical question answering, Knowledge Graph reasoning, memory-aware interaction, polyp segmentation quality, and Polyp VQA performance. Through these experiments, the thesis examines how modular agent coordination affects response quality, grounding, latency, and safety behavior in a medical assistant setting.

## 1.5 Contributions

The main contributions of this thesis are organized around architecture, grounding, and evaluation:

- a modular multi-agent architecture that separates routing, text reasoning, vision reasoning, memory retrieval, and human validation while allowing intermediate evidence to be shared across agents;
- a grounded Text Module that coordinates multi-turn clarification, query expansion, RAG, Knowledge Graph reasoning, MedGemma CoT sufficiency assessment and answer synthesis, and fallback search to reduce unsupported or premature medical responses;
- a Vision Module that connects general medical image analysis, polyp segmentation, and segmentation-grounded Polyp VQA for multimodal reasoning;
- a polyp segmentation pipeline with connected-component post-processing and region-wise color dominance refinement to improve mask consistency for interpretation and downstream VQA grounding;
- a grounded Polyp VQA formulation that adapts MedGemma-4B-it and Qwen3.5-4B with LoRA and compares original-image with mask-guided inputs;
- a patient-scoped long-term memory layer using Mem0 to support contextual continuity while keeping stored patient-reported facts separate from verified diagnoses;
- an evaluation design that analyzes both component performance and system behavior using Dice Score, BLEU-4, ROUGE-L, LLM Judge, One-Hop Accuracy, Multi-Turn Awareness Score, and memory-oriented scenarios.

## 1.6 Research Questions

This thesis is guided by five research questions:

- How can a medical assistant combine conversational reasoning, RAG, Knowledge Graph reasoning, and Web Search to reduce unsupported medical responses?
- How can information sufficiency assessment inside the MedGemma CoT workflow improve safety in symptom-based consultation?
- Can segmentation masks improve Polyp VQA by providing visual grounding for a fine-tuned VLM?
- How can long-term memory support personalization without turning patient-reported facts into verified diagnoses?
- What trade-offs appear between modularity, response quality, latency, and safety in a multi-agent medical assistant?

These questions connect the architecture with the evaluation methodology. The thesis therefore evaluates the system as a reasoning architecture and not as a clinical product ready for deployment.

## 1.7 Scope of the Thesis

The thesis focuses on AI methods and system-level reasoning rather than user-interface engineering. Its main contribution lies in agent orchestration, knowledge retrieval, multimodal reasoning, long-term memory integration, and validation-oriented safety.

The medical scope is also limited. The system is intended to provide information support and preliminary reasoning. It does not replace physicians, does not provide definitive diagnosis, and does not make treatment decisions. For high-risk symptoms or uncertain findings, the system should recommend professional medical consultation.

# CHAPTER 2. BACKGROUND AND RELATED WORK

## 2.1 Large Language and Vision-Language Models in Healthcare

Large language models have become a central direction in medical AI because they can represent clinical knowledge, answer medical questions, and generate patient-facing explanations [17]. Med-PaLM and Med-PaLM 2 show that instruction tuning, medical-domain adaptation, and prompting strategies can substantially improve medical question answering [18]. Open medical LLMs such as PMC-LLaMA and BioMistral follow another direction by adapting general-purpose models to biomedical literature, textbooks, and medical instruction data [19, 20]. These studies show clear progress, but they mostly evaluate the model as an independent QA system rather than as part of a complete consultation workflow.

Medical vision-language models extend this trend from text-only reasoning to image-text reasoning. LLaVA-Med and Med-Flamingo adapt multimodal foundation models for biomedical image conversation, VQA, and rationale generation [21, 22]. More recent models such as MedGemma provide medical vision-language backbones, while general model families such as Qwen offer strong multilingual and instruction-following capabilities that can be adapted to medical tasks [51, 52]. In gastrointestinal endoscopy, datasets such as Kvasir-VQA and Kvasir-VQA-X1 further indicate a shift from image classification toward question answering over clinical visual evidence [44, 45].

However, current LLMs and VLMs still have important limitations for medical assistants. They can generate fluent but unsupported answers, especially when the user question lacks clinical context or when the model relies only on parametric memory [23]. Many models also operate without explicit retrieval, Knowledge Graph evidence, segmentation grounding, or human validation. For VLMs, correct-looking answers may come from dataset priors or textual shortcuts rather than from the relevant image region, which is risky in endoscopy because lesion interpretation depends on local boundary, texture, morphology, and color patterns.

This thesis therefore treats LLMs and VLMs as components inside a larger safety-aware architecture. The proposed system combines model generation with routing, RAG, Knowledge Graph reasoning, MedGemma CoT-based information sufficiency assessment and answer synthesis, patient-scoped memory, segmentation-based visual grounding, and human validation. This design addresses the limitation of standalone models by testing whether external evidence and spatial grounding improve reasoning reliability rather than only answer fluency.

## 2.2 Medical Question Answering and Multi-hop Reasoning

Medical question answering has traditionally been evaluated through benchmark datasets such as PubMedQA, MedQA, and MedMCQA [24, 25, 26]. These datasets are valuable because they test biomedical reading comprehension, medical exam knowledge, and multiple-choice clinical reasoning. Vietnamese medical QA datasets such as VimedAQA further show the importance of language-specific resources for local healthcare applications [13]. However, these benchmarks usually provide a well-formed question and a fixed answer space. Real users often provide incomplete, informal, or ambiguous descriptions, so benchmark performance alone is not sufficient to evaluate a consultation-oriented assistant.

The key difference is that medical consultation is frequently underspecified at the first turn. A user asking about chest pain, abdominal pain, fever, dizziness, or medication use may omit onset time, severity, duration, associated symptoms, age, pregnancy status, allergies, chronic diseases, or current medication. A direct answer may therefore be clinically incomplete even if it is linguistically fluent. This creates a gap between single-turn medical QA and interactive medical reasoning.

Multi-hop reasoning addresses this gap by decomposing a complex question into intermediate reasoning steps and evidence requirements. General multi-hop QA research shows that answering some questions requires connecting multiple supporting facts rather than retrieving one isolated passage [27]. In the medical domain, the same principle appears in two forms. The first form is evidence multi-hop reasoning, where the system combines information from multiple sources such as retrieved documents, Knowledge Graph relations, and previous conversation context. The second form is multi-turn clarification, where the system asks follow-up questions before generating a final response.

This thesis focuses on multi-turn clarification as a safety mechanism and uses evidence multi-hop reasoning for KG/RAG evidence synthesis. The Conversation Agent helps identify missing clinical variables, while KG and RAG agents construct structured and unstructured evidence when the query is answerable from available sources. MedGemma CoT then acts as an information-sufficiency decision point: it decides whether the system should synthesize an answer, ask a targeted follow-up question, or activate fallback search. This workflow makes clarification more than a prompting technique; it becomes a control mechanism for deciding when to answer, when to retrieve, when to ask, and when to recommend professional care. It also separates general medical education from patient-specific advice: a question such as “What is gastritis?” can often be answered from retrieved explanatory material, whereas “Do I have gastritis?” requires additional clinical variables and a more cautious response.

## 2.3 Retrieval-Augmented Generation for Medical QA

Retrieval-Augmented Generation combines neural generation with external document retrieval [14]. Following the formulation of Lewis et al., the generator is not expected to answer only from parametric memory; it is conditioned on retrieved passages that provide non-parametric knowledge for the current input. In a medical assistant, this design is useful because answers can be grounded in retrieved medical documents instead of relying only on the model's internal weights. A typical RAG pipeline includes query expansion, embedding-based retrieval, document reranking, and evidence-conditioned answer synthesis.

The proposed system uses RAG for Vietnamese medical question answering. Documents are embedded and stored in a vector database, while MedGemma CoT uses retrieved context to produce structured, safety-aware answers. This approach is useful when the answer requires explanatory text, treatment descriptions, disease overviews, or patient-friendly medical information.

RAG is not sufficient by itself. Retrieval can fail when a query is underspecified, when relevant documents are absent from the corpus, or when semantically similar but clinically irrelevant passages are retrieved. A retrieved passage can also make an answer appear grounded even when the original clinical question remains incomplete. For this reason, the proposed Text Module combines RAG with query expansion, reranking, context filtering, Knowledge Graph reasoning, and MedGemma CoT sufficiency assessment and answer synthesis.

Medical RAG also requires attention to language and domain mismatch. In Vietnamese medical QA, users may mix colloquial expressions with formal medical terms. A retrieval system must bridge this gap without retrieving unrelated passages. Query expansion and filtering are therefore not auxiliary features; they are necessary parts of making retrieval usable in real consultation-style interactions.

## 2.4 Medical Knowledge Graphs

Knowledge Graphs represent medical knowledge as entities and relations. In the conventional graph view, nodes describe entities and edges encode typed relationships between them; in a medical setting, these entities may include diseases, symptoms, drugs, body parts, risk factors, tests, or treatments. Relations encode structured links such as disease-has-symptom, drug-treats-disease, or disease-associated-with-risk-factor.

Compared with RAG, a Knowledge Graph is more suitable for structured reasoning. It can support explicit traversal of disease-symptom relationships and provide more interpretable evidence for certain diagnostic suggestions. In this thesis, the Knowledge Graph Agent queries VietMedKG through Neo4j and supports reasoning for symptom-to-disease and related medical questions [4].

Knowledge Graphs also have limitations. They depend on relation coverage and graph construction quality. A graph may encode important disease-symptom relations but lack detailed explanation or updated guidance. Therefore, this thesis treats KG and RAG as complementary rather than competing approaches.

The complementarity is especially relevant for medical reasoning. A Knowledge Graph can provide a compact relation such as a symptom-disease link, but a patient-facing answer often requires explanation, uncertainty, and practical next steps. RAG can provide that explanatory material. Conversely, RAG may retrieve long text but fail to make an explicit relation clear. KG can provide structure to support the final synthesis. In the proposed system, these two evidence channels are not treated as final answers; they are passed to MedGemma CoT, where sufficiency, uncertainty, safety boundaries, and final response content are handled together.

## 2.5 Multi-agent AI Systems

Multi-agent systems divide complex tasks among specialized agents. In medical AI, this is useful because different queries require different capabilities. A symptom consultation may require multi-turn clarification and sufficiency assessment, a disease explanation may require RAG, a structured reasoning query may require Knowledge Graph traversal, an image question may require a vision-language model, and a follow-up question may require patient memory.

The proposed system uses a Decision Agent to route user requests to the appropriate agents. The workflow is orchestrated with LangGraph [1], which allows the system to maintain short-term state, coordinate agent execution, retrieve patient memory before routing, and route to fallback strategies when internal knowledge is insufficient.

The multi-agent design also improves observability. The system can identify which agent handled a query and which source of information was used. This is useful for debugging, evaluation, and user trust in safety-sensitive medical applications.

However, multi-agent systems introduce coordination challenges. Routing errors, inconsistent outputs between agents, and increased latency can reduce system quality. A medical multi-agent assistant must therefore include clear routing rules, fallback behavior, and evaluation of intermediate decisions. This thesis treats agent orchestration as a core technical contribution rather than as an implementation detail.

## 2.6 LoRA and Parameter-Efficient Fine-tuning

Fine-tuning large VLMs is computationally expensive because full-parameter updates require significant memory and training resources. LoRA provides a parameter-efficient alternative by freezing the pretrained model weights and learning low-rank adaptation matrices that are injected into selected layers [50]. This reduces the training cost and makes domain adaptation more practical.

In this thesis, LoRA is used to adapt VLMs for Polyp VQA. The goal is to improve the model's ability to answer endoscopy-related visual questions while keeping fine-tuning feasible under limited resources. This is appropriate for the thesis setting because endoscopic VQA data is specialized and smaller than the generic multimodal corpora used to train foundation models.

The fine-tuning experiment is designed around two compact base models: MedGemma-4B-it and Qwen3.5-4B. MedGemma is a medical vision-language model based on the Gemma family and is designed to provide stronger medical image and text understanding than a general-purpose model of similar scale [51]. Qwen models provide a strong multilingual and instruction-following foundation, with Qwen3 extending the family toward hybrid reasoning and broader multilingual coverage [52]. Comparing these two backbones is useful because Polyp VQA requires both visual understanding and language generation: MedGemma contributes domain-specific medical pretraining, while Qwen provides a strong general instruction-following baseline.

LoRA also makes it practical to compare input formulations. In this work, Polyp VQA is evaluated with original-image input and segmentation-grounded input, allowing the analysis to separate the effect of model adaptation from the effect of spatial grounding. The segmentation-grounded setting uses the mask as a prior that indicates where the model should inspect the original image, rather than as a substitute for visual interpretation.

## 2.7 Polyp Segmentation and Neoplasm Characterization

Colonoscopy polyp segmentation is a key medical image analysis task because accurate localization of polyps can support colorectal cancer screening and clinical interpretation. The task can be binary, where the model separates polyp from background, or multi-class, where the model also distinguishes neoplastic and non-neoplastic regions. Compared with general semantic segmentation, polyp segmentation is difficult because polyps vary greatly in size, color, texture, shape, and boundary clarity. The boundary between polyp tissue and surrounding mucosa can be ambiguous, and artifacts such as blur, specular highlights, low contrast, and imperfect bowel preparation can further reduce visual quality [38].

Public datasets have strongly shaped research in this area. Kvasir-SEG provides manually annotated gastrointestinal polyp images with segmentation masks and is widely used for benchmarking binary polyp segmentation [39]. For neoplasm characterization, the BKAI-IGH NeoPolyp dataset extends the problem from localization to multi-class segmentation, where the model must distinguish neoplastic and non-neoplastic regions [40]. This makes the task closer to clinical interpretation than a pure foreground-background mask, but it also increases class imbalance and label ambiguity.

Early deep learning approaches for biomedical segmentation were strongly influenced by U-Net, whose encoder-decoder structure combines contextual feature extraction with precise localization through skip connections [35]. U-Net++ further redesigned skip pathways with nested dense connections to reduce the semantic gap between encoder and decoder features [36]. These architectures remain important baselines because medical datasets are often small and require models that can learn from limited annotated data.

DeepLabV3+ introduced another influential direction by combining atrous convolution, Atrous Spatial Pyramid Pooling, and a decoder for boundary refinement [37]. Its design follows the common semantic-segmentation goal of preserving stronger object-level context while recovering sharper boundaries in the decoder. The multi-scale receptive fields are useful for polyp segmentation because lesions can occupy very different image areas. This thesis follows the DeepLabV3+ family for the deployed segmentation model and produces a three-class output, while the quantitative comparison is reported later in the experimental results.

Attention-based and transformer-based methods address different limitations. PraNet uses a parallel partial decoder and reverse attention to refine boundary cues and recover missed regions [38]. Polyp-PVT replaces a CNN backbone with a pyramid vision transformer and introduces modules for cascaded fusion, camouflage identification, and similarity aggregation, improving robustness in challenging cases such as small objects and appearance variation [41]. More recent medical segmentation work also emphasizes efficient decoders; EMCAD proposes a multi-scale convolutional attention decoder that reduces parameters and FLOPs while maintaining strong performance across medical segmentation tasks [42].

Foundation segmentation models introduce another research direction. Polyp-SAM adapts the Segment Anything Model to the polyp domain and shows that foundation models can be transferred to medical segmentation, but domain adaptation remains necessary because generic segmentation priors do not automatically solve endoscopic boundary ambiguity [43]. This supports the broader design principle of the thesis: general foundation models are useful starting points, but task-specific adaptation and evaluation are still required for medical workflows.

In this thesis, segmentation is not only an end result but also an intermediate artifact for multimodal reasoning. A clean mask can help the Polyp VQA model focus on clinically relevant regions, while an incorrect mask can misdirect the answer. Therefore, segmentation quality, visual grounding, and VQA quality must be evaluated together rather than as isolated components.

## 2.8 Long-Term Memory for AI Agents

Most dialogue agents maintain short-term state within a session, but medical interaction often requires continuity across sessions. Long-term memory allows an agent to remember clinically useful facts such as allergies, chronic diseases, medication use, repeated symptoms, and patient preferences.

In the proposed system, Mem0 is used as a patient-scoped long-term memory layer backed by vector retrieval [8, 9]. The Mem0 design emphasizes persistent agent memory: salient facts are extracted from interactions, stored with embeddings and metadata, and retrieved when they are relevant to a later query. In this thesis, that idea is adapted conservatively for medical use. Mem0 does not replace LangGraph, RAG, the Knowledge Graph, or patient data storage. Instead, it provides additional context that helps agents personalize responses, support routing, and avoid repeatedly asking for information that was already provided.

Medical memory must be handled conservatively. A stored fact may represent a user-reported allergy or symptom history, but it should not be treated as a verified diagnosis. Memory is therefore used as supporting context rather than authoritative clinical truth, and it should remain subordinate to the current user input, retrieved medical evidence, and current image findings.

# CHAPTER 3. PROPOSED ARCHITECTURE OF THE MULTI-AGENT MEDICAL ASSISTANT

## 3.1 System Overview and Orchestration

### 3.1.1 Overall Architecture

The proposed system is organized as a multi-agent medical assistant with four main layers:

- **Decision and orchestration layer:** selects the appropriate processing path;
- **Text Module:** handles medical QA, multi-turn consultation, RAG, Knowledge Graph, and fallback search;
- **Vision Module:** handles general medical image interpretation, polyp segmentation, and Polyp VQA;
- **Memory and validation layer:** provides long-term patient context and human-in-the-loop feedback.

This architecture is designed to reduce reliance on a single model. Instead of asking one LLM to solve every task, the system delegates work to agents that are specialized for retrieval, graph reasoning, visual interpretation, segmentation, or VQA.

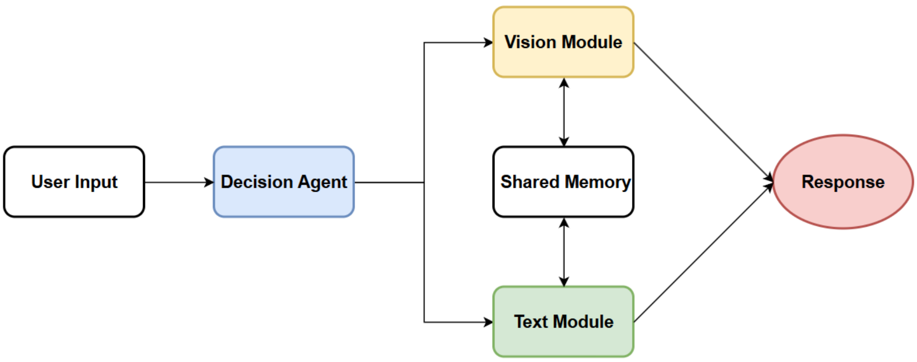


**Figure 3.1:** Overall architecture of the Multi-Agent Medical Assistant

Figure 3.1 shows that the system is not organized as a linear chatbot pipeline. The Decision Agent first interprets the modality and intent of the user input, then routes the request to the Text Module or Vision Module. Shared memory is placed between these modules rather than attached to only one agent. This design allows text-based reasoning and image-based reasoning to exchange context while still keeping their internal models and evidence sources separate.

The high-level architecture in Figure 3.1 establishes the main separation of responsibilities. The more detailed workflow in Figure 3.2 explains how these responsibilities are coordinated during execution and is used below to analyze the system benefits.

### 3.1.2 LangGraph-based Agent Orchestration

LangGraph is used as the workflow orchestration framework. It maintains state across the processing flow and connects decision nodes with task-specific agents. The graph-based structure is useful because medical consultation is not a simple single-step process. A query may require routing, follow-up questioning, retrieval, fallback search, image analysis, validation, or memory update.

The short-term state includes the current query, image path when available, selected agent, conversation history, patient identifier, and intermediate outputs such as segmentation paths or retrieved contexts. This state supports coherent routing and answer synthesis within the current session.

The graph structure also makes the workflow easier to control than a purely prompt-based pipeline. Each node has a clear role: routing, retrieval, image analysis, MedGemma CoT reasoning, validation, or memory update. This is useful in a medical context because the system can inspect intermediate outputs before producing the final answer. For example, if retrieval fails or the image route is uncertain, the workflow can move to a fallback path instead of forcing a response from incomplete evidence.

Another advantage is that LangGraph separates short-term state from long-term memory. Short-term state is useful for the current interaction, while Mem0 stores facts that may remain useful across sessions. This separation prevents temporary reasoning artifacts, such as a retrieved passage or a provisional model output, from automatically becoming persistent patient memory.

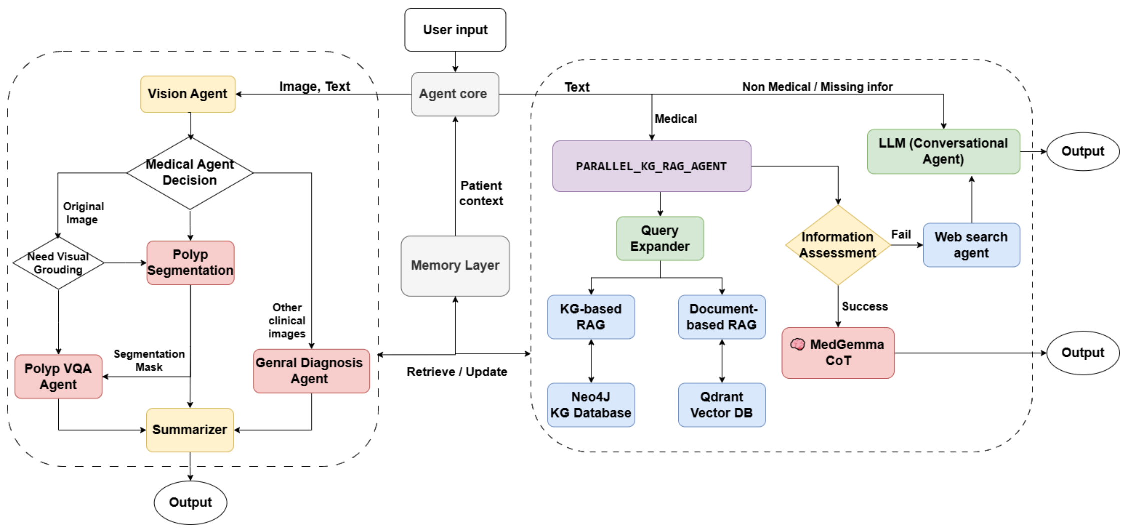


**Figure 3.2:** Detailed agent workflow with text, vision, memory, and fallback coordination

Figure 3.2 shows both the system components and the routes between them. The Agent Core separates text-oriented and image-oriented requests while receiving patient context from the Memory Layer. In the text branch, medical queries pass through parallel KG/RAG evidence construction, query expansion, KG-based RAG, document-based RAG, and MedGemma CoT synthesis. If the selected evidence is weak or the user query is missing important context, the workflow can move to Web Search or conversational handling instead of forcing a final medical answer.

The image branch in Figure 3.2 routes image-text inputs through a vision decision step. Colonoscopy-related tasks can use polyp segmentation and, when needed, segmentation-grounded Polyp VQA. Other clinical images are routed to the general visual diagnosis path. A summarizer then converts visual-agent outputs into a user-facing response. Table 3.1 summarizes the architectural benefits that follow from this workflow.

| Benefit | How Figure 3.2 supports the benefit |
|---|---|
| **Functional decomposition** | User input is routed from the Agent Core to specialized text and vision branches. This means heterogeneous medical tasks can be handled by agents with suitable tools instead of forcing one model to perform retrieval, reasoning, segmentation, VQA, summarization, and fallback handling at the same time. |
| **Evidence specialization** | KG, RAG, Web Search, shared memory, and segmentation masks appear as separate evidence channels. The system can therefore combine structured relations, explanatory documents, updated web evidence, patient context, and visual grounding without collapsing them into one opaque prompt. |
| **Graceful fallback** | MedGemma CoT evaluates the selected KG/RAG context and routes weak-evidence cases to Web Search or conversational handling. This helps avoid premature answers when internal evidence is missing, uncertain, or clinically underspecified. |
| **Cross-modal continuity** | Vision outputs and text reasoning interact through shared memory and workflow state. A user can upload an image, receive an initial interpretation, and then continue asking follow-up questions that reuse the image-derived context. |
| **Observability and evaluation** | Intermediate modules such as routing, query expansion, KG/RAG retrieval, segmentation, visual grounding, summarization, and MedGemma CoT reasoning are explicit. Errors can therefore be traced to a specific stage, and the system can be evaluated beyond the final generated answer. |

**Table 3.1:** Benefits of the multi-agent design illustrated by the detailed workflow

The table clarifies that the system is not merely a collection of independent models. Its value comes from how intermediate evidence is produced, assessed, reused, and routed before MedGemma CoT produces the final response.

### 3.1.3 Agent Cooperation and Cross-module Workflows

A key characteristic of the proposed system is that agents do not operate as isolated modules. They cooperate through three shared objects: **workflow state**, **long-term memory**, and **intermediate evidence**. This means that the output of one agent can become structured input for another agent.

The cooperation mechanism can be summarized as follows:

- **Text cooperation:** Conversation Agent, Query Expansion, KG Agent, RAG Agent, Web Search Agent, and MedGemma CoT cooperate to decide whether to ask, retrieve, reason, or fall back.
- **Vision cooperation:** Polyp Segmentation Agent produces a lesion mask that becomes visual grounding for the Polyp VQA Agent, while the summarizer consolidates visual-agent outputs into the final response.
- **Cross-modal cooperation:** image-derived context can support later text follow-up questions, while text context can guide image interpretation.
- **Memory-supported cooperation:** patient-reported facts can help routing and personalization, but they remain supporting context rather than confirmed clinical evidence.

| **Cooperation pattern** | **Agents involved** | **Shared evidence** | **Main benefit** |
|---|---|---|---|
| **Text reasoning** | Conversation, Query Expansion, KG, RAG, MedGemma CoT, Web Search | Follow-up needs, expanded query, KG facts, retrieved passages, fallback evidence | Safer medical QA |
| **Segmentation-to-VQA** | Polyp Segmentation Agent, Polyp VQA Agent, and Summarizer | Segmentation mask, overlay image, and visual-agent answer | Better visual grounding and clearer final response |
| **Cross-modal follow-up** | Vision agents, Conversation Agent, Text Module | Image result, short-term state, user follow-up question | Interactive consultation |
| **Memory-assisted routing** | Memory Layer, Decision Agent, task-specific agents | Patient-reported allergies, medication use, chronic conditions, prior findings | Continuity across sessions |

**Table 3.2:** Main cooperation patterns in the proposed multi-agent architecture

**Text reasoning pattern.** The Text Module uses cooperation to avoid premature answers. The Conversation Agent first checks whether the user has provided enough clinical context. If the query is sufficiently specified, Query Expansion reformulates it into medically useful terms. The KG Agent then returns structured relations such as symptom–disease links, while the RAG Agent retrieves explanatory documents. These evidence sources are passed into MedGemma CoT, which evaluates information sufficiency and decides whether to synthesize an answer, ask a follow-up question, or activate fallback search.

**Segmentation-to-VQA pattern.** In the Vision Module, the Polyp Segmentation Agent produces a mask that localizes clinically relevant regions in a colonoscopy image. This mask is not only a final segmentation output; it is reused as a visual grounding signal for the Polyp VQA Agent when the task requires visual grounding. The VQA Agent can then answer natural-language questions using both the original image and the segmentation overlay, and the Summarizer converts the visual outputs into a concise user-facing response.

**Cross-modal follow-up pattern.** After a user uploads a medical image, the interaction can continue as a dialogue. The user may ask what a highlighted region means, whether a finding is concerning, or what symptoms should be monitored. The system keeps image-derived context in short-term workflow state and routes later text questions back to the relevant visual evidence or to the Text Module for medical explanation.

**Memory-supported pattern.** Shared memory provides continuity across sessions. For example, previous patient-reported facts can help the Decision Agent route a new medication question more carefully. However, memory is not treated as a diagnosis. It supports context construction, while current symptoms, retrieved evidence, and image findings remain the primary basis for reasoning.

Overall, agent cooperation turns modularity into an evidence-passing workflow. Segmentation contributes spatial precision, VQA provides language-based interpretation, KG contributes structured relations, RAG provides explanatory grounding, and the Conversation Agent manages missing information. The architecture is therefore valuable because specialized agents produce intermediate evidence that other agents can inspect, reuse, and evaluate.

### 3.1.4 Decision Agent

The Decision Agent is responsible for determining which agent should handle a request. For text-only queries, it decides whether the query should be handled by Conversation Agent, KG/RAG reasoning, or Web Search. For image queries, it routes to General Diagnosis Agent, Polyp Segmentation Agent, or Polyp VQA Agent.

The decision process considers four main signals:

- **Modality:** whether the input is text-only, image-only, or image with a question;
- **Intent:** whether the user asks for explanation, symptom consultation, segmentation, VQA, or fallback search;
- **Conversation context:** whether the current query refers to previous turns or uploaded images;
- **Patient context:** whether long-term memory provides relevant allergies, medications, chronic conditions, or prior findings.

| Routing case | Preferred route | Reason for this route |
|---|---|---|
| **General or incomplete conversation** | Conversation Agent | Collect missing details or answer non-specialized dialogue safely. |
| **Medical text query with enough context** | KG/RAG reasoning and MedGemma CoT | Retrieve evidence, compare sources, and synthesize a cautious medical response. |
| **Weak internal evidence** | Web Search fallback or follow-up question | Avoid forcing an answer when internal sources do not support the query. |
| **General medical image** | General Diagnosis Agent | Provide cautious visual interpretation without assuming a polyp-specific workflow. |
| **Colonoscopy segmentation request** | Polyp Segmentation Agent | Produce a spatial mask for direct inspection and downstream grounding. |
| **Polyp image question** | Polyp VQA Agent | Answer an image-question pair using the specialized VQA pipeline. |

**Table 3.3:** Decision Agent routing behavior

Routing errors are important because they can create downstream failures. If a symptom consultation is routed directly to final synthesis, the system may skip necessary follow-up questions. If a general image is routed to the specialized polyp workflow, the segmentation model may produce meaningless output. If a polyp VQA question is routed to a generic VLM, the answer may lack domain-specific reasoning. Therefore, the Decision Agent is evaluated not only as a convenience component but also as a safety-critical part of the architecture.

**Conservative routing principle.** When the task is ambiguous, the system should prefer clarification or a safer general path over a highly specialized agent. Specialized agents are useful only when the intent and input type match their assumptions.

### 3.1.5 Knowledge and Context Sources

The system combines several knowledge and context sources. They are not equally reliable, and they should not be merged blindly. Table 3.4 summarizes the role and limitation of each source.

| Source | Primary contribution | Main limitation |
|---|---|---|
| **RAG documents** | Vietnamese explanatory medical text stored in a vector database. | May retrieve text that is semantically similar but clinically weak or too general. |
| **Knowledge Graph** | Structured disease, symptom, drug, and relation evidence. | May be incomplete or too sparse for patient-friendly explanation. |
| **Web Search** | Fallback coverage for missing or updated information. | Requires careful source filtering and more conservative claims. |
| **Short-term state** | Current conversation context and intermediate workflow outputs. | Useful only within the current session and may include provisional reasoning. |
| **Long-term memory** | Patient-reported facts retrieved through Mem0. | May be outdated, self-reported, or incomplete. |
| **Visual masks** | Spatial grounding for Polyp VQA. | Depends on segmentation quality and may propagate mask errors. |

**Table 3.4:** Knowledge and context sources with reliability considerations

This combination allows the system to support both explanatory medical answers and structured reasoning. The final response should reflect the strength and uncertainty of each source rather than treating all retrieved context as equally authoritative.

### 3.1.6 Safety and Human-in-the-loop Validation

The system includes safety-oriented response behavior. It provides medical disclaimers, recommends professional consultation when necessary, and avoids presenting AI output as a definitive diagnosis. The human-in-the-loop validation mechanism allows users to confirm or comment on generated responses. This feedback can support later quality review and can help determine which facts are reliable enough to be stored in long-term memory.

- **Before generation:** routing, retrieval, and sufficiency checks reduce unsupported answers.
- **During generation:** prompts require uncertainty, warning signs, and care guidance.
- **After generation:** human validation collects user feedback for review and memory governance.

Human validation is placed after the AI response rather than inside every internal reasoning step. This keeps the system usable while still allowing feedback to be collected. In the scope of this thesis, validation is treated as a safety and review mechanism, not as proof of clinical correctness. The system remains a decision-support tool and should not be interpreted as replacing professional medical judgment.

## 3.2 Text Reasoning Module

### 3.2.1 Role and Processing Flow

The Text Module is responsible for medical reasoning over user-provided language. It handles questions about symptoms, diseases, drugs, medical concepts, and follow-up concerns. Its main goal is to produce MedGemma CoT responses that are grounded, cautious, and aware of missing clinical information.

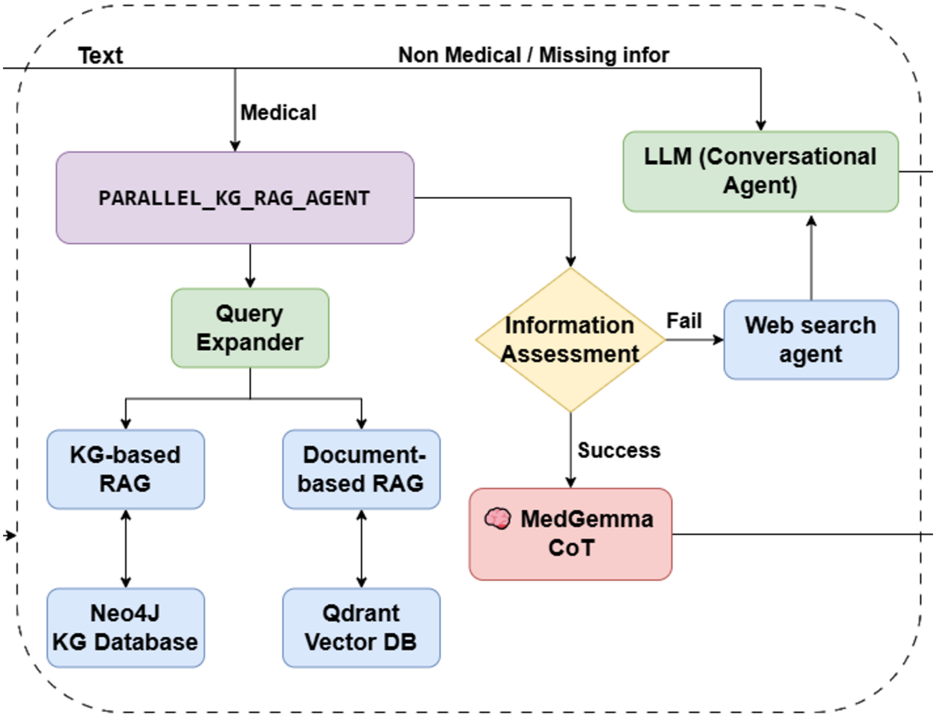


**Figure 3.3:** Text Module architecture for medical knowledge retrieval

Figure 3.3 provides the high-level view of the Text Module. It shows that text reasoning begins with user input classification, then branches into conversational handling, KG/RAG evidence retrieval, MedGemma CoT assessment and synthesis, and fallback search. Although the workflow diagram may visualize assessment as a separate decision node, this node should be interpreted as a logical step inside MedGemma CoT rather than as an independent agent. This figure is therefore introduced at the beginning of the module section before the individual agents are discussed in detail.

**Core idea.** The Text Module is not only a retrieval component. It is a control layer that decides when to ask, when to retrieve, when to synthesize evidence, and when to avoid a patient-specific conclusion.

The general processing flow begins with input analysis, then moves to clarification when the query is clinically underspecified, evidence retrieval when the query is answerable from medical sources, and MedGemma CoT assessment and synthesis when candidate evidence is available. This flow is designed to make the Text Module cautious by default. The module does not assume that every query should immediately trigger a final answer. Instead, it separates the tasks of understanding the user intent, checking information completeness, retrieving evidence, and producing a safe final answer through MedGemma CoT.

The Text Module therefore acts as the main reasoning layer for non-image medical queries. It must handle two very different types of user input. The first type is educational, such as asking what a disease is or how a medication works. The second type is consultation-oriented, such as describing symptoms and asking what might be happening. The second type requires stricter information sufficiency checks because the same symptom can have different meanings depending on duration, severity, age, medical history, and associated signs.

The workflow in Figure 3.2 also shows that text reasoning is not reduced to document retrieval alone. The Text Module combines conversational intake, KG/RAG context selection, MedGemma CoT reasoning and synthesis, and web fallback. This layered structure is important because the main risk in medical QA is not only whether the system can retrieve facts, but whether the reasoning component knows when the available context is insufficient for a patient-specific conclusion.

### 3.2.2 Text Query Types and Routing Behavior

The Text Module is designed around the observation that medical text queries are not homogeneous. Different query types require different agent behavior. A general educational question can usually be answered after retrieval, while a patient-specific symptom question may require follow-up questioning before retrieval. A medication-related question may require checking contraindications, allergy context, and source reliability. A non-medical or socially conversational input should not be forced into a medical reasoning pipeline.

| Query type | Main agent behavior | Reasoning requirement |
|---|---|---|
| **Medical education** | Use RAG and/or KG to retrieve reliable medical knowledge. | Provide grounded explanation without over-personalizing the answer. |
| **Symptom consultation** | Pass user context to MedGemma CoT for sufficiency assessment and ask follow-up questions if needed. | Avoid premature diagnosis when onset, severity, duration, risk factors, or warning signs are missing. |
| **Structured relation query** | Route to KG reasoning and use RAG for explanation when needed. | Connect entities such as symptoms, diseases, drugs, body parts, and risk factors. |
| **Medication or treatment question** | Use retrieval cautiously and consider patient context such as allergies or chronic diseases when available. | Avoid unsafe recommendations and encourage professional consultation for dosage or treatment decisions. |
| **Missing or non-medical input** | Use the conversational agent or request clarification. | Prevent irrelevant retrieval and keep the system aligned with user intent. |

**Table 3.5:** Text query types and corresponding routing behavior

Table 3.5 shows why routing is part of the reasoning problem. If every text query is treated as a retrieval query, the system may return medical documents even when the user actually needs clarification. Conversely, if every symptom query is treated as a conversational query, the system may underuse available medical evidence. The Text Module therefore uses routing to decide when to ask, when to retrieve, when to reason over graph relations, and when to synthesize a response.

### 3.2.3 Conversation Agent, Multi-turn Awareness, and KG-based Reasoning

The Conversation Agent handles cases where the user's query is incomplete or conversational. In medical consultation, missing information is common. For example, a symptom description may lack onset time, severity, location, duration, associated symptoms, medical history, or current medication use.

In this thesis, the term multi-hop has two related but distinct meanings. At the dialogue level, the system performs **multi-turn clarification**: it collects missing information step by step before producing a patient-specific response. At the evidence level, the KG-based reasoning branch can follow structured relations such as symptom–disease, disease–treatment, disease–medication, and disease–advice. The former is evaluated as multi-turn awareness, while the latter is discussed as an architectural advantage of combining Knowledge Graph evidence with RAG.

Multi-turn clarification is used when the current user input is not sufficient for a safe answer. Instead of generating a diagnostic suggestion immediately, the agent asks focused follow-up questions. This process helps the system approximate a structured clinical intake conversation. The goal is not to replace a doctor, but to reduce the risk of giving an answer from insufficient evidence.

The follow-up questions are designed to be concise and clinically relevant. For symptom-based consultation, the system prioritizes red flags and high-impact missing variables. For example, in a fever-related query, duration, peak temperature, associated respiratory or gastrointestinal symptoms, medication use, and risk factors may change the response. In a pain-related query, location, severity, onset, radiation, and accompanying symptoms are important.

This behavior improves usability because the system does not overwhelm the user with a full medical form. It asks only the next useful question based on the current context.

The multi-turn design also improves traceability. Each follow-up question corresponds to a missing clinical variable, such as onset, duration, severity, location, associated symptoms, medication use, allergy history, or risk factors. This makes it possible to analyze whether the agent is asking useful questions or merely extending the conversation. In evaluation, this behavior is reflected by the Multi-Turn Awareness Score, which rewards the system for recognizing when additional information is needed before answering.

The practical effect of this behavior is analyzed in the Text Module results section, where the traditional single-turn pattern is compared with the proposed multi-turn clarification pattern.

#### 3.2.3.1 Illustrative Multi-turn Consultation Flow

An illustrative symptom consultation demonstrates why multi-turn behavior is needed. Suppose the user writes: “I have stomach pain and nausea. What disease could it be?” A single-turn chatbot may immediately list gastritis, food poisoning, reflux, or appendicitis. However, this answer is clinically weak because the initial input lacks duration, pain location, severity, fever, vomiting, stool changes, medication use, pregnancy status, and warning signs.

In the proposed workflow, the Conversation Agent first treats the query as underspecified. Instead of giving a strong conclusion, it asks a focused follow-up question such as: “Where is the pain located, how long has it lasted, and do you have fever, vomiting blood, black stool, severe pain, or persistent vomiting?” This question is not arbitrary; it targets variables that can change the safety framing of the response. If the user later reports mild epigastric pain after meals without red flags, the system can retrieve information about gastritis or reflux and provide cautious advice. If the user reports severe right-lower abdominal pain with fever, the system should prioritize urgent professional care rather than ordinary health education.

This example illustrates the difference between direct answering and consultation support. The goal of the Conversation Agent is not to prolong the dialogue, but to determine whether the current evidence is sufficient for the next safe action. The final response may still include possible causes, but it should clearly separate likely explanations, missing information, warning signs, and next steps.

### 3.2.4 Clinical Context Requirements before MedGemma CoT

Information sufficiency is not implemented as a standalone agent. It is a decision step inside MedGemma CoT that receives two inputs from the upstream Text Module: the user’s clinical context and evidence retrieved by KG/RAG agents. KG and RAG provide structured relations, candidate diseases, matched symptoms, and explanatory passages; MedGemma CoT decides whether this context is enough to answer safely.

**Sufficiency is a decision boundary.** A query can have retrieved evidence and still be clinically insufficient. The system should not confuse “documents were found” with “the user gave enough context for patient-specific advice.”

This distinction is necessary because retrieved evidence does not always mean that the user’s case is clinically clear. The system should distinguish between:

- general medical education questions that can be answered from knowledge sources;
- symptom-based questions that require follow-up details;
- high-risk cases that should recommend immediate professional care;
- out-of-scope or non-medical queries.

In implementation, sufficiency is treated as a dynamic threshold. It is lower for general education questions and higher for patient-specific recommendations, medication-related questions, or symptoms associated with urgent conditions. The result changes the route of execution: the system may synthesize an answer, ask the Conversation Agent to collect missing details, recommend urgent care, or invoke

Web Search when KG/RAG context is weak.

This design is also reflected in evaluation. Multi-Turn Awareness Score measures whether MedGemma CoT recognizes missing clinical context and triggers follow-up behavior rather than forcing an answer. The metric should therefore be interpreted as awareness of insufficient context, not as direct accuracy of path-level KG multi-hop reasoning.

#### 3.2.5 Medical Knowledge Retrieval with RAG and Knowledge Graph

The system uses both RAG and Knowledge Graph retrieval because they provide complementary strengths.

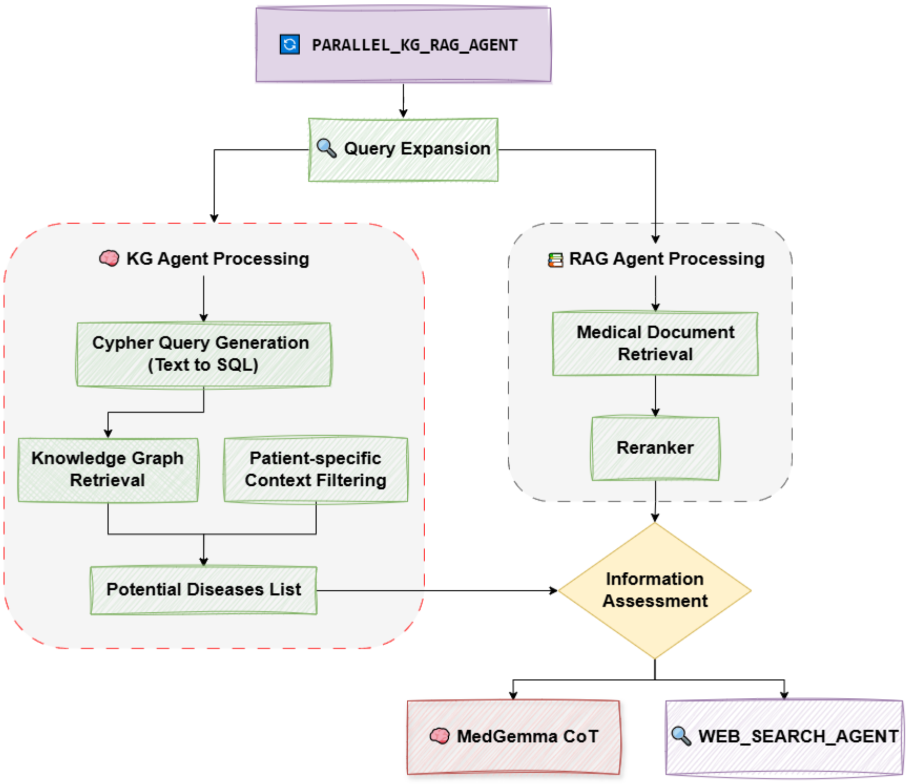


**Figure 3.4:** Parallel KG-RAG processing in the Text Module

The parallel design in Figure 3.4 reflects a central assumption of this thesis: unstructured and structured medical knowledge are complementary. RAG is strong when the system needs explanatory evidence, while KG reasoning is strong when the question depends on explicit relations between entities such as symptoms, diseases, drugs, and body parts. Running these branches in parallel allows the system to compare different evidence forms before generating an answer.

##### 3.2.5.1 Data Retrieval Layer

The data retrieval layer is responsible for collecting candidate evidence before medical reasoning begins. In the proposed Text Module, retrieval is not limited to one database. The RAG branch retrieves unstructured medical documents from the Qdrant vector database, while the KG branch retrieves structured disease and symptom relations from Neo4j [3, 4]. This dual-source design allows the system to use both semantic similarity and explicit medical relations.

Before retrieval, Query Expansion reformulates the user question by adding related medical terms, synonyms, anatomical terms, and possible disease names. This is especially important for Vietnamese medical queries because users may describe symptoms in informal language, while the data source may use more formal medical terminology. For example, a user may say "đau bao tử", while relevant documents may use terms such as epigastric pain, gastritis, reflux, or stomach pain.

The vector retrieval branch uses dense embedding search. In the implementation, documents are embedded with `google/embeddinggemma-300m` and stored in Qdrant using cosine similarity [12, 3]. Dense retrieval is useful because it can match semantically related expressions even when the exact keywords differ. The retrieval design can also be extended with hybrid search, where dense vector search is combined with keyword-based retrieval such as BM25 [34]. This is useful in medical QA because some terms, drug names, disease names, and abbreviations require exact lexical matching, while symptom descriptions often benefit from semantic matching.

After candidate documents are retrieved, a reranker reorders them according to query-document relevance [33]. This step is important because initial vector similarity may return documents that are semantically close but clinically less useful. The reranker helps prioritize documents that better match the actual user question before MedGemma CoT receives the selected context.

**Retrieval interpretation.** RAG is most useful for explanatory medical text, while Knowledge Graph retrieval is most useful for structured relationships. The system uses both because medical QA often needs a relation-level hypothesis and a patient-friendly explanation.

#### 3.2.5.2 KG Agent

The KG Agent retrieves structured medical knowledge from Neo4j [4]. Its main role is to convert natural-language medical questions into graph queries and then use the retrieved relations as structured evidence. This is different from RAG because the KG branch does not only search for similar text; it traverses explicit relations between entities such as diseases, symptoms, treatments, medications, advice, and patient-related attributes.

The Cypher Query Service is responsible for translating user questions into Cypher queries. In the implementation, this process is supported by `GraphCypherQAChain` and few-shot examples [6]. Few-shot prompting is useful because medical graph queries often require a stable query pattern: identify the target entity, traverse relevant relations, and return disease candidates or associated attributes. A direct LLM answer may be fluent but unsupported, while a Cypher query forces the reasoning step to retrieve concrete graph evidence.

The KG Agent also supports patient-specific context integration. Retrieved graph results can be filtered using patient information such as age, gender, medical history, current diseases, and symptom context. This prevents the system from returning graph results that are structurally related but clinically less relevant to the patient.

For example, a disease relation may exist in the graph, but it may be less appropriate if it mainly concerns a different age group or conflicts with the patient's reported history.

Context filtering can use embedding similarity between the patient context and disease descriptions. The filtered results are then ranked as disease candidates, including information such as matched symptoms, total symptoms, and similarity scores. This makes the KG output more than a raw list of relations: it becomes a patient-aware candidate set that can be inspected by MedGemma CoT before final synthesis.

The main advantage of KG-based reasoning is that the system can organize evidence as explicit medical relations rather than as isolated text snippets. For example, a symptom cluster can lead to candidate diseases, and each candidate can then be connected to treatment methods, medication information, prevention advice, or associated conditions. This supports multi-hop evidence construction such as symptom $\rightarrow$ disease $\rightarrow$ treatment or symptom $\rightarrow$ disease $\rightarrow$ warning signs. In the architecture, this capability is treated as a structured evidence pathway rather than as a final answer by itself.

#### 3.2.5.3 RAG Agent

The RAG Agent retrieves and processes unstructured medical documents, mainly from VimedAQA through Qdrant [13, 3]. Its purpose is to provide explanatory medical context that can be used in patient-facing responses. While the KG Agent is strong at structured relation reasoning, the RAG Agent is better suited for definitions, disease descriptions, general advice, medication explanations, and practical health information.

The RAG branch follows three main steps. First, Query Expansion improves the input query by adding relevant medical terms and synonyms. Second, dense retrieval searches the vector database for top-$k$ semantically related passages. Third, the reranker sorts the retrieved passages by practical relevance to the user query [33]. This reranking step helps reduce noise before MedGemma CoT synthesizes an answer.

RAG also improves transparency because the final answer can be grounded in retrieved documents rather than only in the parametric memory of the LLM. However, retrieved documents should still be assessed critically. A passage can be medically valid but irrelevant to the patient's specific situation. Therefore, the RAG output is passed as selected context into MedGemma CoT instead of being used blindly.

The KG and RAG branches therefore serve different but complementary purposes. KG contributes compact relational structure and candidate hypotheses, while RAG contributes explanatory context and patient-facing language. The contribution of the Text Module is therefore not choosing KG or RAG in isolation, but coordinating them according to the evidence need and information sufficiency of each query. The comparative behavior of these alternatives is reported later in the experimental results.

#### 3.2.5.4 Coordinated KG-RAG Evidence Construction

The KG and RAG branches contribute different forms of evidence. The KG branch is useful for identifying candidate relations, such as a disease being associated with a symptom or a drug being related to a disease category. The RAG branch is useful for converting those relations into explanatory language that can be understood by non-expert users. This distinction matters because a medical assistant must both reason and communicate.

For example, when the user asks about a symptom cluster, the KG branch can return candidate diseases connected to the symptoms. These candidates should not be treated as diagnoses. Instead, they serve as structured hypotheses that guide retrieval and MedGemma CoT synthesis. The RAG branch can then retrieve documents explaining the candidate conditions, typical symptoms, warning signs, and general management. MedGemma CoT can synthesize these sources into a cautious answer: possible explanations, why they may be relevant, what information is missing, and when to seek medical care.

This coordination also reduces two opposite risks. If the system relies only on KG, the answer may become too sparse and relation-like, lacking patient-friendly explanation. If the system relies only on RAG, the answer may become verbose and less structured, especially when retrieved passages contain overlapping or weakly relevant information. By combining both, the Text Module can use KG for structure and RAG for explanation.

| Stage | Agent output | Role in final response |
|---|---|---|
| **Query Expansion** | Medical synonyms, related symptoms, disease names, and anatomical terms. | Bridges colloquial user language and formal medical terminology. |
| **KG Agent** | Structured entity relations and candidate disease-symptom links. | Provides compact reasoning structure and candidate hypotheses. |
| **RAG Agent** | Retrieved explanatory passages from medical documents. | Provides patient-friendly explanation and supporting context. |
| **MedGemma CoT** | Candidate analysis, red-flag screening, information sufficiency assessment, and final answer synthesis over selected KG/RAG context. | Determines whether to answer, ask follow-up questions, use fallback search, or produce a cautious response without presenting it as a definitive diagnosis. |

**Table 3.6:** Coordinated evidence construction in the Text Module

### 3.2.6 Parallel Reasoning and Fallback Strategy

The system can retrieve and filter RAG and Knowledge Graph results in parallel. These agents do not make the final medical decision; they select context for MedGemma CoT. If the selected internal context is insufficient after CoT assessment, the system can use Web Search as a fallback or ask the user for missing information. In the implementation, Web Search fallback is backed by Tavily search tooling [10].

This strategy reduces dependence on any single source. RAG may fail if the relevant document is not retrieved, while the Knowledge Graph may fail if the needed relation is missing. If one internal source is weak, the other source can still provide partial support. If both internal sources are weak, Web Search becomes a controlled fallback rather than an uncontrolled first response. Parallel retrieval does not mean that all sources are merged blindly: KG/RAG agents first select and filter context based on relevance, patient context, and retrieval quality, and MedGemma CoT then compares source consistency and clinical sufficiency.

#### 3.2.7 MedGemma CoT for Medical Decision Support and Answer Synthesis

After KG and RAG agents select candidate context, the MedGemma CoT component performs evidence synthesis and decision support. Web Search is treated as a fallback source when internal context is weak or incomplete, not as the default first source. The role of MedGemma CoT is not to create a definitive diagnosis, but to organize available evidence into a clinically cautious reasoning process. This component receives candidate diseases, matched symptoms, retrieved passages, patient context, and the current conversation state.

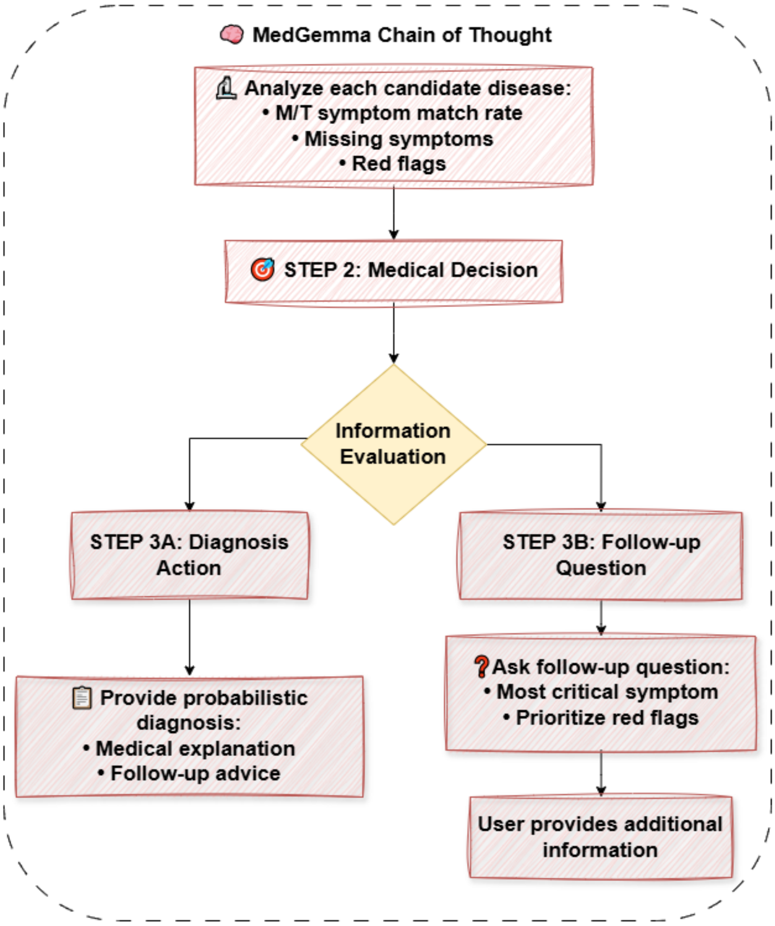


**Figure 3.5:** Medical chain-of-thought, information sufficiency, and answer synthesis flow

Figure 3.5 summarizes the decision flow of the MedGemma CoT component: it analyzes candidate diseases, checks red flags, evaluates information sufficiency, and then either produces a cautious diagnostic suggestion or asks follow-up questions.

The first step is candidate-level analysis. For each candidate condition, the model compares the user's reported symptoms with the symptom profile retrieved from KG/RAG. This includes estimating whether the candidate explains the main complaint, which symptoms are matched, which symptoms are missing, and whether the available evidence is too weak. This step is important because a candidate disease should not be listed only because it appears in a retrieved document or graph relation.

The second step is red-flag screening. MedGemma CoT checks whether the conversation contains warning signs such as severe pain, difficulty breathing, neurological symptoms, bleeding, persistent vomiting, high fever, chest pain, or other urgent indicators depending on the symptom category. If red flags are present, the system should prioritize escalation and professional consultation instead of continuing ordinary explanation.

The third step is medical decision branching. If the retrieved evidence and user context are sufficient, the system can synthesize a cautious diagnostic suggestion with likely explanations and monitoring advice. If the evidence is insufficient, the system should activate multi-turn clarification and ask the next clinically useful follow-up question. This branch is central to the safety design: the system is allowed to defer the answer when the information is incomplete.

| CoT step | Input considered | Output behavior |
|---|---|---|
| **Candidate analysis** | Disease candidates, matched symptoms, retrieved passages, and patient context. | Compare possible conditions and identify which candidates are plausible, weak, or unsupported. |
| **Red-flag screening** | Reported symptoms, severity, duration, urgent signs, and risk factors. | Escalate to urgent-care advice when dangerous patterns appear. |
| **Information assessment** | Completeness of clinical variables and consistency of evidence sources. | Decide whether to answer, ask follow-up questions, or use fallback search. |
| **Answer synthesis** | Filtered evidence, candidate ranking, and safety constraints. | Produce a cautious explanation, possible causes, missing information, and next-step guidance. |

**Table 3.7:** MedGemma CoT decision-support stages

This design connects retrieval with safe MedGemma CoT synthesis. KG and RAG provide evidence, but MedGemma CoT decides how that evidence should be interpreted under uncertainty. A high-ranking candidate can still be rejected if key symptoms are missing, and a low-information query can trigger follow-up questioning even when retrieval returns many documents. Therefore, the MedGemma CoT stage prevents the system from confusing evidence retrieval with clinical sufficiency.

When MedGemma CoT decides that enough information is available, its final output should:

- answer the user's question directly when possible;
- explain the reasoning in understandable language;
- cite or mention sources when available;
- state uncertainty when information is incomplete;
- recommend professional care for severe or urgent symptoms;
- avoid presenting the AI output as a definitive diagnosis.

The generated response is then available for human-in-the-loop validation. The final answer is therefore the result of a controlled MedGemma CoT reasoning process rather than a raw model completion. The response format should be adapted to risk level: low-risk educational questions can receive concise explanations; symptom-based questions should include known information, missing information, warning signs, and next steps; urgent or red-flag cases should prioritize escalation over detailed explanation.

MedGemma CoT should also preserve the provenance of reasoning as much as possible. If the answer is mainly based on RAG evidence, it should emphasize retrieved medical information. If the answer depends on KG relations, it should explain that the listed conditions are related possibilities rather than confirmed diagnoses. If Web Search is used, the response should be more conservative because source quality may vary. This provenance-aware response style improves user trust and makes the system easier to audit.

In addition, the response should separate *medical knowledge* from *patient-specific interpretation*. For example, the system can explain that gastritis commonly causes epigastric pain and nausea, but it should not conclude that the user has gastritis without sufficient context. This separation is central to safe medical assistance: the assistant may provide useful education and triage guidance, but it should avoid replacing clinical diagnosis.

## 3.3 Vision Reasoning Module

### 3.3.1 Role and Processing Flow

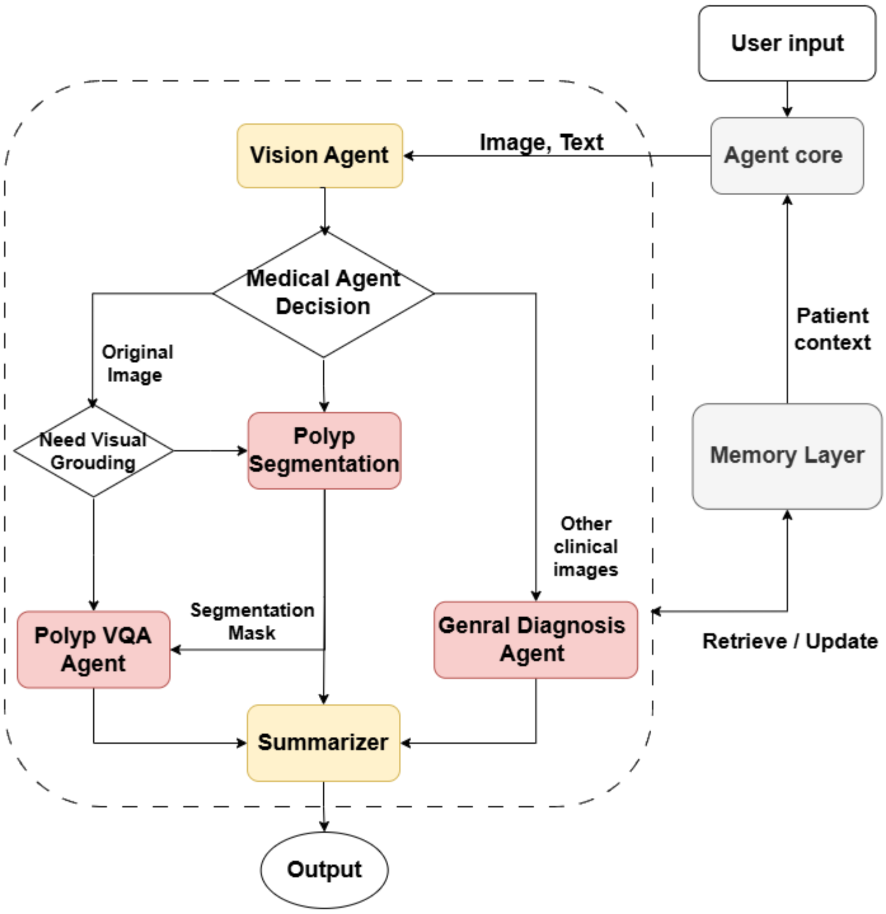


**Figure 3.6:** Vision Module workflow with routing, optional grounding, summarization, and memory context

The Vision Module processes image-based medical inputs. It supports general medical image interpretation and a specialized colonoscopy polyp workflow. The module is designed to coordinate visual models with the multi-agent architecture so that image findings can be combined with text questions and patient context.

The module follows a task-specific design. General medical image interpretation is kept broad and cautious, while polyp-related processing is more specialized and more formally evaluated. This allows the system to support multiple image inputs without treating all image types as equally reliable.

The design intentionally separates general image analysis from specialized polyp analysis. A general VLM may be useful for describing visible patterns, but colonoscopy polyp segmentation and Polyp VQA require domain-specific assumptions, datasets, and evaluation metrics. This separation makes the system more transparent: the user and evaluator can identify whether an answer comes from a general visual model or from the specialized polyp workflow.

Figure 3.6 expands the image branch of the full workflow. The Agent Core receives image-text input and can use patient context from the Memory Layer before routing to the Vision Agent. The medical image decision step then separates polyp-specific tasks from other clinical images. For polyp images, the workflow may use the original image directly for Polyp VQA or first generate a segmentation mask when visual grounding is needed. For other clinical images, the request is routed to the General Diagnosis Agent. The Summarizer combines the selected visual-agent output into the final response, while memory can be retrieved or updated as supporting patient context.

### 3.3.2 Image Routing and Visual Task Classification

When an image is uploaded, the system determines whether the task is general image interpretation, polyp segmentation, or Polyp VQA. If the user explicitly requests mask generation or segmentation, the image is routed to the Polyp Segmentation Agent. If the user asks a question about an endoscopic polyp image, the image is routed to the Polyp VQA Agent, with segmentation added when visual grounding is needed. Other medical images can be handled by the General Diagnosis Agent.

This routing step is important because the same image can support different tasks. A user may ask for a segmentation mask, a description of visible findings, or a question about risk. Routing prevents the system from applying the wrong agent to the wrong task.

Visual routing also depends on the accompanying text. An image without a question may require general description, while the same image with a question about polyp type should trigger Polyp VQA. If the user asks specifically for localization or mask generation, segmentation becomes the primary task. If the question requires evidence from a suspected lesion region, the system can use the segmentation mask as a grounding signal before VQA. The image and the text prompt are therefore interpreted together rather than independently.

### 3.3.3 General Diagnosis Agent

The General Diagnosis Agent handles medical images outside the specialized polyp workflow. It uses a multimodal model to describe visible findings, provide cautious interpretation, and recommend appropriate follow-up when necessary.

This agent is positioned as a support tool rather than a diagnostic authority. Its output should be interpreted as preliminary information and should not replace clinical review by trained professionals.

Because the General Diagnosis Agent covers a broad space of image modalities, it is not the main formal evaluation target in this thesis. Its role is to provide system coverage and demonstrate extensibility of the vision pipeline. The specialized polyp workflow receives more detailed quantitative evaluation.

The agent should avoid unsupported claims about diagnosis, severity, or treatment. Its most appropriate output style is a cautious description of visible content, possible medical relevance, and recommendation for professional interpretation when the image is clinically significant. This safety framing is necessary because different medical image modalities require different expert knowledge and acquisition conditions.

### 3.3.4 Polyp Segmentation Agent

The Polyp Segmentation Agent is designed for colonoscopy images. Its task is to segment polyp regions and distinguish between background, neoplastic polyp regions, and non-neoplastic polyp regions. The segmentation output is useful for two purposes: direct visual interpretation and visual grounding for Polyp VQA.

This agent converts an image-level input into a spatially localized representation. That representation is easier to inspect than a text-only answer and can be passed to other agents. In the proposed architecture, segmentation is therefore both a prediction task and an intermediate reasoning artifact.

The three-class design is also useful for downstream reasoning. A binary mask can indicate where a polyp is located, but it does not distinguish neoplastic from non-neoplastic regions. By preserving class information, the mask can provide richer evidence to the Polyp VQA Agent. This is particularly relevant when the user's question asks about suspicious areas or class-related interpretation.

#### 3.3.4.1 DeepLabV3+ for Polyp Segmentation

DeepLabV3+ is used as the core segmentation architecture for the Polyp Segmentation Agent. The target output is a dense three-class mask in which each pixel is assigned to background, neoplastic polyp region, or non-neoplastic polyp region. This setting requires both accurate localization and stable class prediction inside the lesion area.

**Architecture objective.** DeepLabV3+ is selected because it combines high-level semantic context with boundary recovery. Colonoscopy frames often contain lesions with variable size, weak contrast, specular highlights, and unclear tissue boundaries. A useful architecture must therefore capture multi-scale context while still preserving enough spatial detail for pixel-level prediction.

**Main components.** The architecture contains four important parts:

- **Backbone encoder:** extracts hierarchical feature maps from the input colonoscopy image. The deeper layers provide semantic information, while lower-level features retain edge and texture details.
- **Atrous convolution:** expands the receptive field without excessively reducing feature-map resolution. This is useful when a polyp boundary requires both local texture cues and wider mucosal context.
- **ASPP:** Atrous Spatial Pyramid Pooling applies parallel atrous convolution branches at multiple dilation rates, together with image-level pooling, to capture objects and contextual patterns at different scales.
- **Decoder:** combines upsampled high-level features with low-level features to recover spatial detail and produce sharper segmentation boundaries.

The segmentation agent follows the DeepLabV3+ encoder-decoder design, summarized in Figure 3.7. The encoder first extracts hierarchical features from the input image through a deep convolutional backbone with atrous convolution. Atrous convolution is important because it enlarges the effective receptive field without reducing the feature-map resolution too aggressively. In colonoscopy frames, this is useful because the model must capture both local texture changes around the lesion boundary and broader contextual information from the surrounding mucosa.

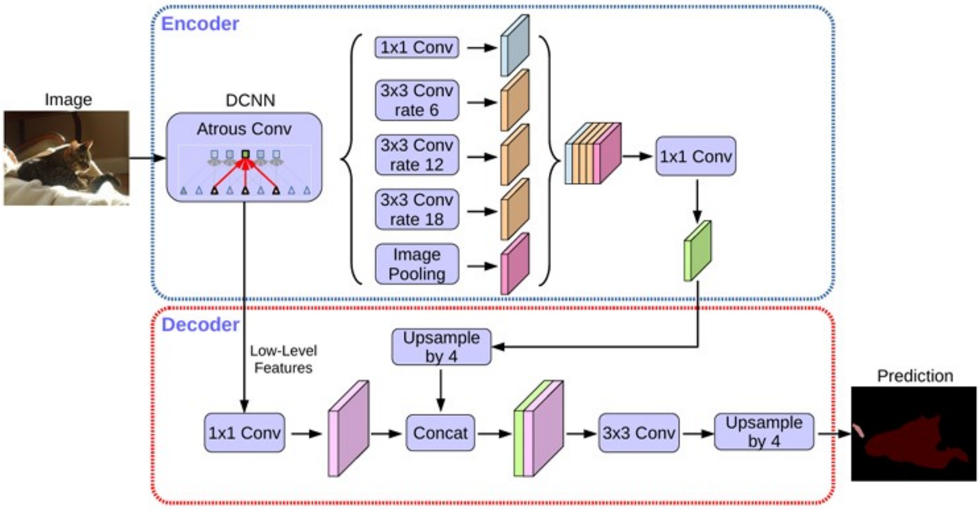


**Figure 3.7:** DeepLabV3+ encoder-decoder architecture with Atrous Spatial Pyramid Pooling

**Processing workflow.** Let $X$ be the input colonoscopy image. The encoder first produces high-level semantic features:

$$F_e = ASPP(DCNN(X)). \tag{3.1}$$

The ASPP output is then upsampled and concatenated with low-level features $F_l$ extracted from earlier layers:

$$F_d = Concat(Upsample(F_e), Conv_{1\times1}(F_l)). \tag{3.2}$$

The decoder refines the combined representation and produces the final pixel-wise prediction:

$$P = Upsample(Conv_{3\times3}(F_d)). \tag{3.3}$$

The dense prediction $P$ is converted into the color mask used by the rest of the system. Blue denotes background, red denotes neoplastic regions, and green denotes non-neoplastic regions. This color convention is not only for visualization; it also provides the symbolic input used by the post-processing method and the segmentation-grounded Polyp VQA setting.

**Relevance to the thesis.** The encoder-ASPP-decoder design matches the needs of three-class polyp segmentation. ASPP helps detect lesions across scale variation, the decoder helps recover boundaries, and the class-wise mask preserves information needed for neoplasm characterization. However, a dense neural prediction can still contain small inconsistent class fragments inside a connected lesion region. The post-processing method below is therefore introduced as a region-level refinement step after DeepLabV3+ prediction.

#### 3.3.4.2 Post-processing and Overlay Mask

Raw segmentation predictions may contain noisy regions or inconsistent labels inside a connected polyp region. To improve consistency, the system applies post-processing based on two main steps.

**Connected Component Analysis** identifies non-background connected regions in the mask. Each connected component is treated as a candidate object region. Let $M(x, y)$ denote the raw predicted color label at pixel $(x, y)$, where blue represents background, red represents neoplastic tissue, and green represents non-neoplastic tissue. The first step converts the predicted mask into a binary region-of-interest map:

$$I(x,y) = \begin{cases} 1, & \text{if pixel } (x,y) \text{ belongs to a target region (non-blue)}, \\ 0, & \text{otherwise.} \end{cases} \tag{3.4}$$

Two pixels $p = (x_1, y_1)$ and $q = (x_2, y_2)$ are considered connected under an 8-neighborhood rule if:

$$|x_1 - x_2| \leq 1, \quad |y_1 - y_2| \leq 1, \quad I(p) = I(q) = 1. \tag{3.5}$$

The connected foreground pixels form a set of connected components:

$$\{R_i\}_{i=1}^{N} \quad \text{such that} \quad \forall (p,q) \in R_i,\ p \leftrightarrow q, \tag{3.6}$$

where $R_i$ is the $i$-th connected region, $N$ is the number of detected connected regions, and $p \leftrightarrow q$ means that there exists a path of connected foreground pixels between $p$ and $q$. The region label map is then defined as:

$$L(x,y) = i \quad \text{if} \quad (x,y) \in R_i. \tag{3.7}$$

**Region-wise Color Dominance Refinement** checks the number of red and green pixels inside each connected component. For each connected region $R_i$, the number of pixels assigned to class $k$ is computed as:

$$C_k(R_i) = \sum_{(x,y) \in R_i} \mathbf{1}[M(x,y) = k], \tag{3.8}$$

where $k \in \{\text{red}, \text{green}\}$ and $\mathbf{1}[\cdot]$ is an indicator function that returns 1 when the condition is true and 0 otherwise. The dominant class of the region is:

$$k^* = \arg\max_k C_k(R_i). \tag{3.9}$$

The same counts also provide a simple measure of how reliable the region-level decision is. Let $|R_i|$ be the number of pixels in region $R_i$. The dominance ratio is:

$$\rho_i = \frac{\max_k C_k(R_i)}{|R_i|}. \tag{3.10}$$

A high $\rho_i$ means that most pixels in the connected region already agree on the same class, so refinement mainly removes small noisy fragments. A low $\rho_i$ indicates that the region is ambiguous because red and green occupy comparable portions. In that case, assigning a single class is more risky and should be interpreted carefully in qualitative analysis.

The refined mask $M'$ is obtained by assigning the dominant class to all pixels in the connected component:

$$M'(x, y) = k^*, \quad \forall (x, y) \in R_i. \tag{3.11}$$

These equations make the assumption of the method explicit. Connected component analysis defines the spatial unit of refinement through $R_i$, while color dominance refinement defines the semantic decision through $C_k(R_i)$ and $k^*$. The method therefore regularizes the existing prediction rather than redrawing the lesion from scratch. The formulation provides three practical insights:

- **Spatial insight from $R_i$:** refinement is performed per connected foreground object, not across the whole image. Therefore, separate candidate lesions are processed independently.
- **Semantic insight from $C_k(R_i)$:** the corrected class is determined by the class distribution already present in the raw prediction. The method regularizes the model output but does not introduce external clinical knowledge.
- **Reliability insight from $\rho_i$:** high-dominance regions are good candidates for refinement, while low-dominance regions should be treated as uncertain cases where the raw prediction itself is class-ambiguous.

After refinement, the mask is overlaid on the original image. The overlay supports human inspection and provides a grounded visual input for the Polyp VQA Agent. This step is useful because a fragmented or mixed-color overlay may guide the VQA model toward inconsistent evidence, whereas a cleaner overlay is easier for both humans and models to interpret. However, the method is intentionally conservative: it improves regional consistency but does not create new lesion regions. Qualitative examples of this behavior are presented in the experimental results.

### 3.3.5 Polyp VQA Agent Design with LoRA Fine-tuned VLMs

The Polyp VQA Agent answers natural-language questions about colonoscopy polyp images. It is invoked when the Decision Agent detects that the user is asking a question about an endoscopic polyp image rather than requesting only segmentation or general image description. The agent receives an image-question input, formats it for the selected VLM endpoint, and returns a concise answer that should be medically relevant and visually grounded in the supplied image.

The agent uses LoRA fine-tuned VLMs hosted through a configurable base URL. In this thesis, the VQA fine-tuning is designed for two 4B-scale backbones: MedGemma-4B-it and Qwen3.5-4B. This design allows the system to compare a medically oriented VLM backbone with a strong general instruction-following backbone while keeping the agent interface compatible with the rest of the workflow.

The dataset construction and experimental split for Polyp VQA are presented in the experimental setup and results chapter. In this architecture chapter, the focus is the agent role: converting image-question inputs into clinically relevant answers and integrating the fine-tuned VLM with the segmentation workflow.

The VQA agent is connected to the segmentation workflow through visual grounding. When a segmentation mask is available, the agent can receive both the original image and the overlay. This design gives the VLM explicit visual evidence about the region of interest while preserving the original image as the primary visual context. If no segmentation mask is available, the agent falls back to the original-image VQA setting.

This design gives the polyp workflow a two-stage reasoning structure: the Segmentation Agent localizes the region of interest, and the Polyp VQA Agent interprets the corresponding visual evidence in language. Segmentation without VQA is difficult for users to interpret, while VQA without segmentation may rely on less explicit visual evidence.

Using hosted fine-tuned VLMs through a base URL keeps the agent workflow independent from the physical deployment of the models. From the agent's perspective, each Polyp VQA model is a callable service that receives an image-question input and returns an answer. The thesis focuses on the AI behavior of these models and their integration with segmentation, not on web backend details.

### 3.3.6 Segmentation-to-VQA Agent Cooperation

The cooperation between the Polyp Segmentation Agent and the Polyp VQA Agent illustrates the role of intermediate visual evidence in the multi-agent architecture. The Segmentation Agent answers a localization question: which pixels correspond to relevant polyp regions, and which predicted class is assigned to those regions? The VQA Agent answers a semantic question: how should the visual evidence be described in response to a natural-language question?

This separation produces the segmentation overlay as an intermediate evidence object. The overlay provides spatial grounding, improves interpretability, and connects pixel-level prediction with language-level reasoning. The VQA Agent does not need to infer the region of interest only from global image features; it receives an explicit visual cue generated by the segmentation pipeline.

| Agent | Main contribution | Limitation when used alone |
|---|---|---|
| **Polyp Segmentation Agent** | Produces pixel-level localization and class-colored masks for neoplastic or non-neoplastic regions. | A mask alone does not explain findings, answer user questions, or provide natural-language clinical reasoning. |
| **Polyp VQA Agent** | Converts image-question input into a clinically relevant natural-language answer. | Without grounding, the VLM may rely on global context, dataset priors, or visually irrelevant regions. |
| **Combined workflow** | Uses the segmentation mask as visual evidence for VQA reasoning. | The workflow depends on mask quality; an incorrect mask can mislead the VQA response. |

**Table 3.8:** Complementary roles of the Polyp Segmentation Agent and Polyp VQA Agent

Table 3.8 summarizes the complementarity between the two agents. The Segmentation Agent improves the precision of visual localization, while the VQA Agent improves the interpretability of the output. Their cooperation is especially relevant for questions such as whether a highlighted region is suspicious, where the polyp is located, or whether the visible region appears consistent with a neoplastic or non-neoplastic class label. These questions require both spatial evidence and language reasoning.

The cooperation also changes how the system should be evaluated. A segmentation model may achieve a high Dice Score but still produce overlays that are confusing for VQA, while a VQA model may produce fluent answers without using the highlighted region. Therefore, this thesis evaluates both the individual models and the segmentation-grounded VQA setting where the mask becomes part of the VQA input.

The results section later evaluates whether this design choice improves answer quality on the mask-matched Polyp VQA subset. In this architecture chapter, the important point is the information flow: segmentation produces spatial evidence, and VQA uses that evidence during answer synthesis.

However, this cooperation is conditional rather than automatically beneficial. If the Segmentation Agent misses the lesion, highlights irrelevant mucosa, or assigns an incorrect class to the region, the VQA Agent may produce a more confident but wrong answer. This is a typical risk of pipeline-based reasoning: an upstream error can propagate downstream. For this reason, the segmentation overlay should be interpreted as an evidence cue, not as unquestionable ground truth. The VQA response should remain cautious, especially when the mask is small, fragmented, uncertain, or inconsistent with the visible image.

#### 3.3.6.1 Example Interaction Pattern

An example workflow illustrates the interaction between the two agents. First, the user uploads a colonoscopy image and asks whether there is a visible polyp. The Decision Agent routes the request to the polyp workflow. The Segmentation Agent generates a mask and overlay, which identifies the suspected region. The Polyp VQA Agent then receives the original image, the overlay, and the user's question. Instead of answering only from the raw image, it can refer to the highlighted region and provide a response such as: the image contains a region that should be reviewed, the highlighted area may correspond to a polyp-like structure, and a clinician should confirm the finding.

The user may then ask a follow-up question: "Why is this region suspicious?" In this case, the system can reuse the same image-derived context. The follow-up question is textual, but it still depends on visual evidence. The workflow can route the question back to the Polyp VQA Agent or combine the VQA output with the Text Module for explanatory medical information. This illustrates cross-turn and cross-modal reasoning: the image is not treated as a one-time input, but as part of the ongoing consultation context.

This interaction pattern is useful for user-facing medical assistance because patients rarely ask only one complete question. They may upload an image, ask what it shows, ask whether the highlighted region is dangerous, ask what symptoms should be monitored, and ask when to see a doctor. The multi-agent workflow supports this behavior by preserving intermediate evidence and allowing later questions to reuse it.

### 3.3.7 Architectural Mechanism of Grounded Polyp VQA

#### 3.3.7.1 Input Formulation and Spatial Prior

The input of the segmentation-grounded VQA agent is not limited to a global endoscopic image. Instead, it is formulated as a composite signal that combines the original endoscopic image $I_{\mathrm{orig}}$, a binary segmentation mask $M_{\mathrm{bin}} \in \{0,1\}^{H\times W}$, and the natural-language question $q$. The VQA input is therefore represented as:

$$X_{\mathrm{vqa}} = (I_{\mathrm{orig}}, M_{\mathrm{bin}}, q)\,. \tag{3.12}$$

In this formulation, $M_{\mathrm{bin}}$ functions as an ROI prior. Pixels with $M_{\mathrm{bin}}(x,y) = 1$ define the spatial support of the suspected lesion region, while pixels with $M_{\mathrm{bin}}(x,y) = 0$ are treated as background with respect to the grounding signal. The mask therefore provides a hard-attention prior: it explicitly constrains the spatial search space before clinical reasoning begins. Rather than allowing the VQA model to infer the relevant image region only from global visual context, the system supplies a geometric prior that indicates where visual evidence should be inspected.

This spatial prior is especially relevant in colonoscopy VQA because clinically meaningful structures often occupy only a small portion of the frame. Polyp boundaries, mucosal texture, surface irregularity, color contrast, and protrusion patterns may be local and visually subtle. By introducing $M_{\mathrm{bin}}$, the system establishes a concrete spatial coordinate reference for the VQA process, reducing the risk that the model answers from irrelevant background tissue or dataset-level visual priors.

#### 3.3.7.2 Task Decoupling through Pure Spatial Grounding

The use of a binary mask also creates a semantic-agnostic spatial reference. In this design, localization and clinical reasoning are separated into two different responsibilities. The Segmentation Agent determines the geometric region that should be inspected, while the VQA Agent remains responsible for interpreting the visual appearance of that region in response to the question.

This separation is important because the mask should not be treated as a complete semantic answer. It tells the VQA model where to look, but it does not by itself determine whether the surface is regular, whether the boundary is clear, whether the morphology appears suspicious, or whether the visible texture supports a particular response. For open-ended questions about shape, surface pattern, color, boundary, or lesion appearance, the model must extract and analyze visual features directly from $I_{\text{orig}}$ inside the region constrained by $M_{\text{bin}}$.

Formally, the spatial support induced by the binary mask is:

$$\Omega_{\text{ROI}} = \{(x, y) \mid M_{\text{bin}}(x, y) = 1\}. \tag{3.13}$$

The grounded visual evidence is therefore not the mask itself, but the set of visual features extracted from $I_{\text{orig}}$ under the spatial constraint defined by $\Omega_{\text{ROI}}$. In this sense, $M_{\text{bin}}$ establishes the spatial grounding signal, while $I_{\text{orig}}$ remains the source of appearance-level evidence. The VQA model is forced to combine geometric guidance with direct visual feature extraction, rather than replacing image interpretation with the segmentation output.

#### 3.3.7.3 Optimization Architecture and Late Cross-modal Alignment

The optimization strategy follows the same separation of responsibility. During fine-tuning, the visual encoder is kept frozen and functions as a static feature extractor. Given the original image and the grounded visual representation, the frozen vision encoder produces a stable set of visual tokens:

$$F_{\text{vis}} = \text{ViT}_{\text{frozen}} \left( I_{\text{orig}}, M_{\text{bin}} \right). \tag{3.14}$$

Because the vision backbone is not updated, the visual representation space is not distorted by the relatively small endoscopic VQA dataset. This is useful in the thesis setting because the available grounded samples are limited and the pretrained visual encoder already provides a general-purpose representation of image patches. Task adaptation is therefore concentrated in the language model through LoRA layers.

The grounded reasoning process can be summarized as:

$$\hat{a} = \text{LLM}_{\theta+\Delta_{\text{LoRA}}} \left( q, F_{\text{vis}}, M_{\text{bin}} \right), \tag{3.15}$$

where $\hat{a}$ is the generated answer, $\theta$ denotes the frozen or mostly preserved pretrained LLM parameters, and $\Delta_{\mathrm{LoRA}}$ denotes the trainable low-rank adaptation parameters. The LoRA layers implement parameter-efficient task adaptation by modifying the attention and projection behavior of the LLM while preserving most pretrained parameters.

Cross-modal fusion is therefore performed late, inside the LLM attention layers, rather than by retraining the visual encoder. Textual tokens from the question, visual tokens from $F_{\mathrm{vis}}$, and spatial cues induced by $M_{\mathrm{bin}}$ are aligned during answer generation. This late cross-modal alignment allows the model to integrate the mask as a guidance signal without disrupting the original visual representation learned by the frozen encoder.

Under this formulation, the segmentation mask guides where the model should look, the original image provides the visual evidence that must be interpreted, and the question determines which visual properties are relevant. The grounded VQA setting should therefore be understood as spatially guided multimodal reasoning, not as replacing clinical visual interpretation with a segmentation label.

#### 3.3.8 Original-image and Segmentation-grounded VQA Settings

The Polyp VQA model is studied under two input settings:

**Original image + question.** The model receives only the original colonoscopy image and the user's question. This represents a VQA setting where the model must identify relevant visual evidence by itself.

**Original image + segmentation mask + question.** The model receives both the original image and the segmentation overlay. The mask acts as a visual grounding signal that guides the model toward the polyp region. This setting represents cooperation between the Polyp Segmentation Agent and the Polyp VQA Agent.

Visual grounding can improve answer quality when the mask correctly highlights clinically relevant regions, but it can also introduce risk if the segmentation mask is inaccurate. The comparison between these settings is therefore one of the central Vision Module experiments. The original-image setting measures the VLM's baseline ability to extract relevant evidence from the endoscopic frame, while the segmentation-grounded setting isolates the effect of adding explicit spatial evidence generated by the Segmentation Agent.

## 3.4 Memory Layer and Context Continuity

#### 3.4.1 Role of the Memory Layer

The Memory Layer provides persistent patient context for the multi-agent system. Its purpose is not to perform diagnosis, but to preserve clinically useful facts that may remain relevant across sessions, such as reported symptoms, allergies, medication use, chronic diseases, treatment reactions, warning signs, lifestyle factors, and communication preferences. This design addresses a limitation of ordinary chat history: short-term conversation state is useful inside one session, but it does not provide reliable continuity when the same patient returns with a related question later.

The proposed system uses Mem0 as the external memory management layer. Mem0 is designed around the idea of extracting salient facts from interactions, storing them with embeddings and metadata, and later retrieving relevant memories for a new query [8, 9]. In this thesis, Mem0 is treated as a memory substrate rather than as a medical reasoning model. That distinction matters: remembering that a patient reported an allergy is useful, but it is not the same as verifying the allergy clinically. Mem0 does not replace LangGraph, RAG, Knowledge Graph reasoning, image analysis, or patient database storage. Instead, it provides an additional patient-context channel that can improve routing, personalization, and continuity.

A key design principle is separation of responsibility. LangGraph manages workflow state and agent coordination; RAG and the Knowledge Graph provide external medical knowledge; the Memory Layer stores reusable patient-specific facts; and task-specific agents generate outputs under safety constraints. This separation prevents long-term memory from becoming an uncontrolled medical knowledge base or a hidden diagnostic record. In particular, the system stores memory as reported or source-attributed facts rather than as confirmed clinical truth.

### 3.4.2 Integration Architecture

The implemented architecture wraps Mem0 inside a patient-scoped service rather than calling the library directly from every agent. The service is implemented in `agents/patient_memory_service/memory_service.py`. It lazily constructs a Mem0 `Memory` instance, configures Qdrant as the vector store, attaches a LangChain LLM for fact extraction, attaches a LangChain embedding model, and passes custom medical memory instructions to Mem0 [8, 9, 3, 5]. The service also standardizes metadata for patient scope, session scope, condition type, source, status, and timestamps.

The memory service exposes four main operations:

- **Explicit condition write:** the condition-write endpoint stores a structured clinical fact with no extra inference. This is used when the system already has a fact that should be preserved exactly.
- **Conversation-based write:** `add_conversation` stores chat turns with `infer=True`, allowing Mem0 to extract durable facts from the interaction.
- **Semantic search:** `search` retrieves top-$k$ memories using a natural-language query, patient filters, optional condition filters, and a relevance threshold.
- **Governance operations:** list and delete endpoints support memory review, correction workflows, and patient-scoped deletion.

The service is also exposed through the FastAPI router `/api/patient-memory` [7]. This makes the Memory Layer usable both inside the LangGraph workflow and as a separate service for debugging, inspection, or future administrative tools. In the current application, `app.py` includes this router, while the graph itself calls the Python service directly.

At the orchestration level, the integration follows a retrieval-before-reasoning and write-after-guardrails pattern. First, the LangGraph node `retrieve_patient_memory` searches for relevant patient memory before the Decision Agent selects an agent. The formatted memory context is then inserted into the routing prompt and into downstream agent prompts as supporting context. After task-agent output, human-validation handling, and output guardrails, the `write_patient_memory` node considers whether the current interaction contains durable facts worth storing. The graph ordering is therefore:

`analyze_input` → `retrieve_patient_memory` → `route_to_agent` → `task agent` → `check_validation` → `apply_guardrails` → `write_patient_memory`.

This placement is intentional. Retrieval before routing allows memory to influence agent selection, while writing after guardrails reduces the risk of storing irrelevant, unsafe, or rejected content.

### 3.4.3 Technical Pipeline

The technical pipeline can be formalized as a patient-scoped memory function. Let $p$ denote a patient identifier, $r$ a session or run identifier, $q_t$ the current user query at turn $t$, $H_t$ the short-term conversation history, and $M_p$ the set of stored memories for patient $p$. Each memory item is represented as:

$$m_i = (x_i, e_i, a_i),$$

where $x_i$ is the natural-language memory text, $e_i = \mathrm{Embed}(x_i)$ is its vector representation, and $a_i$ is a metadata dictionary containing source, type, status, timestamps, and provenance fields. For a new query $q_t$, memory retrieval computes:

$$\mathrm{Retrieve}(q_t, p) = \mathrm{TopK}_{m_i \in M_p} \left( \mathrm{sim}(\mathrm{Embed}(q_t), e_i) \right),$$

subject to metadata filters such as `user_id=p`, `agent_id`, `run_id`, `condition_type`, and `status`. This corresponds to the implemented call `memory.search(query, filters, top_k, threshold)`.

**Memory construction.** The system supports two write modes. In explicit condition writing, the application constructs a memory string from a validated structured payload and stores it without LLM inference. In conversation writing, the system submits user and assistant turns to Mem0 and lets the configured LLM extract concise durable facts according to the medical memory policy. In both cases, Mem0 embeds the resulting memory and stores it in Qdrant with metadata and history tracking [8, 3].

**Listing 3.1:** Patient-scoped memory retrieval before agent routing

```
Input: patient_id p, query q, memory_enabled flag, top_k K,
    threshold tau
Output: formatted memory context C and raw memory items R

if memory_enabled is false or q is empty then
    return C = "", R = []
```

```
end if

filters <- {
    user_id: p,
    agent_id: PATIENT_MEMORY_AGENT_ID
}

R <- Mem0.search(
    query = q,
    filters = filters,
    top_k = K,
    threshold = tau
)

C <- []
for each memory item m in first five items of R do
    source <- m.metadata.source or m.metadata.memory_kind
    if source in {ai_image_analysis, assistant, ai} then
        prefix <- "AI-recorded but unconfirmed"
    else
        prefix <- "Patient-reported"
    end if
    append format(prefix, m.text, source, m.score) to C
end for

return C, R
```

Algorithm 3.1 corresponds to the retrieval and formatting helpers in the implementation. A clinically important detail is that retrieved memory is source-labeled before being added to the prompt. AI-derived image analysis is explicitly marked as unconfirmed, while patient-reported information is marked as reported by the patient. This prevents the model from silently converting memory into confirmed diagnosis.

**Listing 3.2:** Post-response memory writing with medical safety constraints

```
Input: graph state S containing patient_id, session_id,
   user input, agent output
Output: updated state S

if memory is disabled, input/output are empty,
   or S.agent_name == NON_MEDICAL_FILTER then
    return S
end if

input_text <- text(S.current_input)
output_text <- text(S.output)

if S.current_input contains a medical image and output_text
    is not empty then
    fact <- "AI-recorded medical image analysis; requires
       clinician confirmation: "
```

```
            + summarize(output_text)
    Mem0.add(
        messages = [{role: user, content: fact}],
        user_id = S.patient_id,
        agent_id = PATIENT_MEMORY_AGENT_ID,
        run_id = S.session_id,
        metadata = {source: ai_image_analysis, validated:
           false},
        infer = false
    )
    return S
end if

messages_to_store <- shortened user turn and, unless it is
   a validation result,
                    shortened assistant turn
if messages_to_store is not empty then
    Mem0.add(
        messages = messages_to_store,
        user_id = S.patient_id,
        agent_id = PATIENT_MEMORY_AGENT_ID,
        run_id = S.session_id,
        metadata = {source: chat, stored_by:
           langgraph_write_patient_memory},
        infer = true
    )
end if

return S
```

Algorithm 3.2 corresponds to the `write_patient_memory` node. It distinguishes image-derived findings from ordinary chat-derived facts. Image findings are stored explicitly with `infer=False` and `validated=False`; chat turns are passed to Mem0 with `infer=True` so that the configured extraction LLM can extract concise durable memories. The memory policy in `config.py` instructs the system to store only durable patient-condition facts, preserve Vietnamese when appropriate, avoid identifiers and secrets, avoid speculative diagnoses, and mark uncertain information as reported or uncertain.

### 3.4.4 Impact on Agent Reasoning

Long-term memory affects the system at the level of control flow and context construction rather than by directly deciding the final medical answer.

**Decision Agent.** Memory can improve routing. The decision prompt contains the current query, recent conversation context, image status, image type, and retrieved long-term patient memory. For example, if the user asks "Can I take this medicine?" and retrieved memory indicates a penicillin allergy, the query should be routed toward medical reasoning rather than casual conversation.

**Conversation Agent.** Memory can reduce repeated questions. If prior memory already contains chronic disease history, medication use, allergy, or lifestyle factors, the agent can ask more targeted follow-up questions instead of repeatedly collecting the same background information.

**KG/RAG Agents.** Memory can personalize retrieval. Patient context can guide query expansion and evidence synthesis, for example by making medication interactions, risk factors, or chronic disease context more salient. However, RAG and Knowledge Graph evidence remain responsible for medical grounding.

**Vision-related agents.** Memory can provide background context, but image evidence must remain primary. If memory indicates a prior history of colon polyps, this may support routing to the polyp workflow. Nevertheless, segmentation masks, VQA outputs, and the current image should remain the main basis for image-specific interpretation.

**Human validation and guardrails.** Memory is inserted before reasoning, but memory writing is performed after output guardrails. This ordering helps avoid storing content from rejected non-medical image requests and keeps memory updates closer to the final system behavior.

### 3.4.5 Safety Constraints and Limitations

Memory is not a diagnosis. Patient-reported facts should be represented as reported information, such as "the user reports..." or "the patient states...". AI-inferred conclusions should be marked as uncertain unless validated. This distinction is particularly important because retrieval can make an old memory appear authoritative even when the original fact was incomplete, outdated, or only self-reported.

The memory layer should not store raw images, base64 image content, API keys, phone numbers, addresses, or unnecessary personally identifiable information. The implemented custom instructions explicitly reject casual chat, unrelated preferences, secrets, raw identifiers, raw image data, speculative diagnoses, assistant disclaimers, and generic medical advice.

Memory can also become outdated or incorrect, so it must be used as supporting context rather than authoritative truth. A medication that was active in one session may be discontinued later; a symptom may resolve; a suspected disease may be ruled out. Future deployment would require stronger governance mechanisms, including explicit user review, correction, deletion, source tracking, recency weighting, confidence marking, and clinician validation workflows. These mechanisms are especially important when memory is used across multiple sessions because continuity improves usefulness only when the system also supports correction and forgetting.

# CHAPTER 4. DATASETS, EXPERIMENTAL SETUP, AND RESULTS

## 4.1 Datasets

### 4.1.1 VimedAQA Dataset

VimedAQA is used as the Vietnamese medical question-answering source for the RAG component [13]. In the local index used by this thesis, the dataset is represented by 44,228 JSONL records across six groups: body part, disease, drug, and medicine-related passages. Each record contains a medical context, title, keyword, topic identifier, source URL, and author metadata. This structure makes the dataset suitable for retrieval-oriented medical QA rather than direct model fine-tuning.

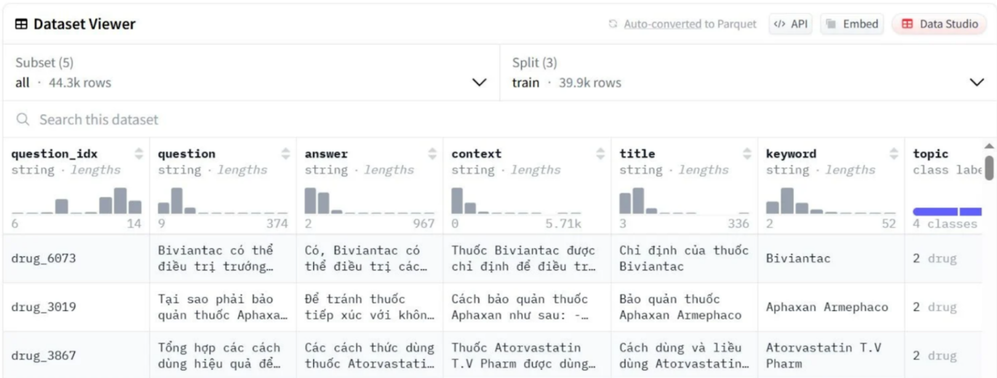


**Figure 4.1:** Medical QA dataset overview used for retrieval-oriented experiments

| VimedAQA Group | Number of Records |
|---|---|
| Body part | 4,956 |
| Disease 1 | 7,500 |
| Disease 2 | 8,154 |
| Drug | 9,754 |
| Medicine 1 | 7,500 |
| Medicine 2 | 6,364 |
| Total | 44,228 |

**Table 4.1:** Local VimedAQA record distribution used for RAG indexing

For retrieval-based reasoning, each record is converted into a retrievable passage. During inference, the user query is embedded and compared with stored passages, and the most relevant contexts are supplied to MedGemma CoT. This setup allows the system to reuse Vietnamese medical information while keeping evidence indexing separate from final answer synthesis.

The quality of this component depends on preprocessing and retrieval design. Noisy passages, duplicated records, or overly broad chunks can reduce retrieval precision. Therefore, the dataset is treated as a knowledge source that requires cleaning, normalization, consistent chunking, and metadata-aware indexing.

### 4.1.2 VietMedKG

VietMedKG is used as the structured medical knowledge source for the Knowledge Graph Agent. In the local data used by this thesis, the graph source table contains 8,900 disease records with fields such as disease name, description, disease type, prevention, causes, symptoms, risk groups, comorbidities, treatment methods, clinical department, tests, nutrition advice, and medication-related information. These fields are converted into structured facts for graph-based retrieval and reasoning.

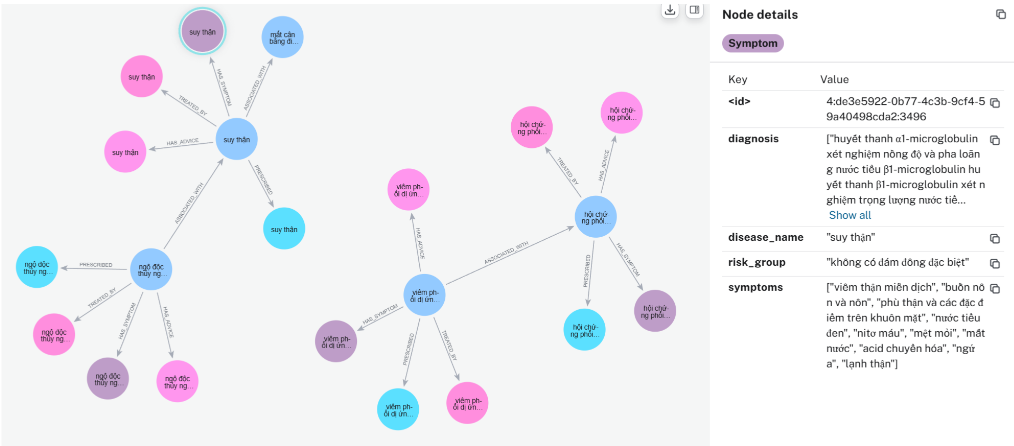


**Figure 4.2:** Overview of the VietMedKG data source used by the Knowledge Graph Agent

VietMedKG complements RAG by supporting relation-oriented reasoning. For example, the system can retrieve symptoms associated with a disease or identify diseases related to a symptom set. This makes the graph particularly useful for one-hop medical relation questions and for checking whether a generated answer is consistent with structured medical facts.

### 4.1.3 BKAI-IGH NeoPolyp-Small Dataset

BKAI-IGH NeoPolyp-Small is used for the polyp segmentation experiments [40]. The dataset contains colonoscopy images with pixel-level annotations for neoplastic and non-neoplastic polyp regions. In this thesis, the segmentation task is formulated as a three-class problem: background, neoplastic polyp region, and non-neoplastic polyp region. This makes the dataset suitable for evaluating both lesion localization and class-aware mask prediction.

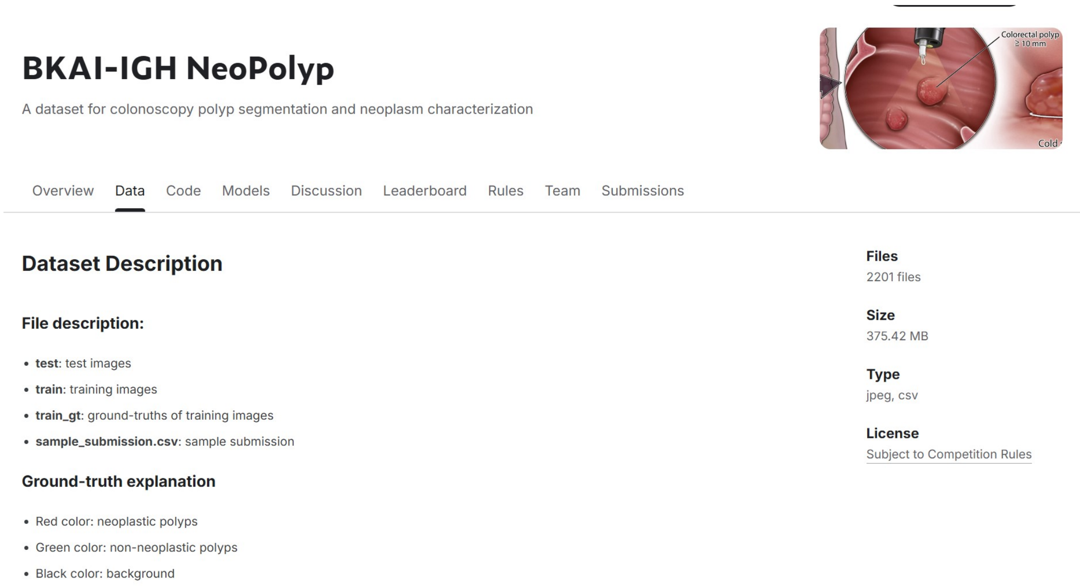


**Figure 4.3:** Example images and annotations from the BKAI-IGH NeoPolyp-Small dataset

The annotations in Figure 4.3 serve as ground truth for Dice Score evaluation and for analyzing whether post-processing improves mask consistency. The three-class setting is important because a mask is not only a geometric localization result; it also carries semantic information that may be used by the VQA agent. Therefore, segmentation errors can affect both direct mask interpretation and downstream visual grounding.

### 4.1.4 Kvasir-VQA-X1 Dataset

Kvasir-VQA-X1 is used as the primary dataset for LoRA fine-tuning and evaluation of the Polyp VQA VLM. It is a gastrointestinal endoscopy visual question answering dataset organized as image-question-answer triples [45]. Compared with classification-only endoscopy datasets, this format asks the model to connect a visual region, a natural-language question, and a clinically meaningful answer. The public dataset extends Kvasir-VQA with 159,549 new question-answer pairs designed for medical reasoning and robustness evaluation. In the local training split used by this thesis, the full VQA setting contains 6,212 original endoscopic images and 143,594 image-question-answer samples.

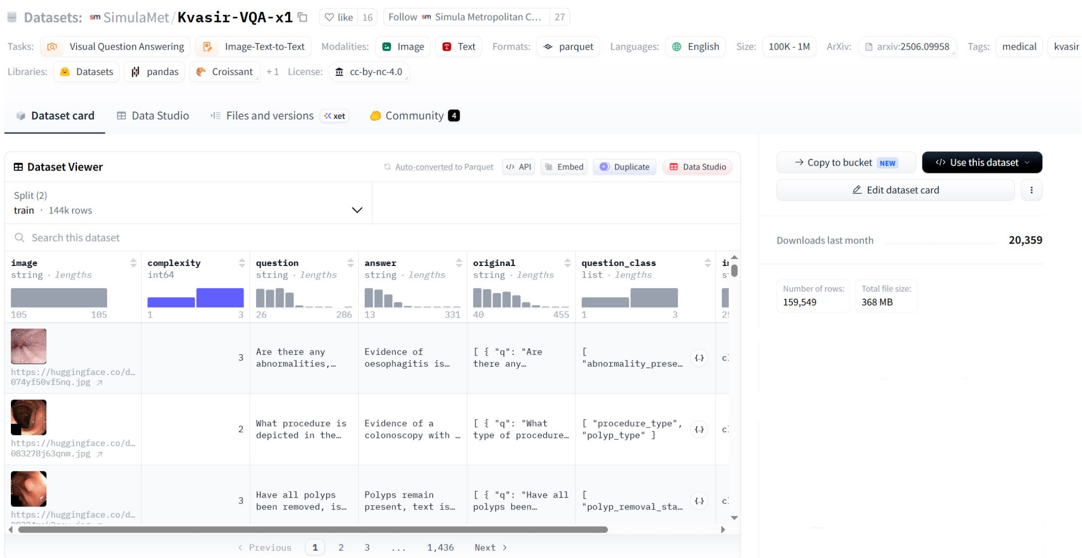


**Figure 4.4:** Example structure of the Kvasir-VQA-X1 dataset for gastrointestinal endoscopy VQA

This structure is suitable for the thesis because the model must connect endoscopic visual evidence with a natural-language question rather than only predict an image label or segmentation mask. Kvasir-VQA-X1 is used in two experimental formats:

- **Original-image VQA:** the model receives the endoscopic image and the question;
- **Visual-grounded VQA:** the model receives the original image, a segmentation mask or overlay, and the question.

The first format evaluates direct visual reasoning from the original frame. The second evaluates whether segmentation-derived grounding improves downstream VQA by giving the model an explicit cue about the region of interest.

The full Kvasir-VQA-X1 training split contains 6,212 original endoscopic images and 143,594 image-question-answer samples. However, segmentation-based visual grounding can only be applied when an image has a matched segmentation mask. After filtering for mask availability, 1,336 images remain with usable segmentation masks, corresponding to 42,382 training samples. Table 4.2 summarizes this filtering step.

| Training Data Setting | Number of Images | Number of Samples |
|---|---|---|
| Original Kvasir-VQA-X1 training data | 6,212 | 143,594 |
| Mask-filtered visual grounding subset | 1,336 | 42,382 |

**Table 4.2:** Kvasir-VQA-X1 training data before and after segmentation-mask filtering

This filtering step is necessary because visual grounding requires both the original image and a corresponding segmentation mask. Therefore, improvements from visual grounding should be compared against a LoRA-only baseline evaluated on the same mask-matched subset, not against a model evaluated on the full dataset. Evaluation data are kept separate from training data to avoid image-level or question-level leakage, and preprocessing is kept conservative to preserve clinically relevant visual evidence.

## 4.2 Experimental Setup and Evaluation Protocol

### 4.2.1 Polyp VQA Fine-tuning Setup

The Polyp VQA models are fine-tuned with LoRA. The training objective is to generate correct and clinically relevant answers from image-question inputs. The fine-tuning setup compares two base-model families, MedGemma-4B-it and Qwen3.5-4B, under both original-image input and visual-grounded input.

Table 4.3 summarizes the main hyperparameters used for LoRA fine-tuning. The same setup is used for MedGemma-4B-it and Qwen3.5-4B in the original-image setting, with the corresponding base model changed according to the experiment.

| Hyperparameter | Original-image LoRA | Visual-grounded LoRA |
|---|---|---|
| Training dataset | Kvasir-VQA-X1 train | Mask-filtered Kvasir-VQA-X1 train |
| Validation dataset | Kvasir-VQA-X1 test-1000 | Mask-matched validation subset |
| Base models | MedGemma-4B-it, Qwen3.5-4B | MedGemma-4B-it |
| Maximum sequence length | 512 | 1024 |
| Torch dtype | bfloat16 | bfloat16 |
| Number of epochs | 1 | 3 |
| Train batch size per device | 4 | 4 |
| Evaluation batch size per device | 4 | 4 |
| Gradient accumulation steps | 4 | 4 |
| Learning rate | $2 \times 10^{-5}$ | $2 \times 10^{-5}$ |
| Learning-rate scheduler | linear | linear |
| Warmup ratio | 0.03 | 0.03 |
| Weight decay | 0.01 | 0.01 |
| LoRA rank | 16 | 16 |
| LoRA alpha | 64 | 64 |
| Freeze vision encoder | true | true |
| Gradient checkpointing | true | true |
| Best-model metric | eval token accuracy | eval token accuracy |
| Checkpoint save interval | 1000 steps | 1000 steps |
| Maximum saved checkpoints | 2 | 2 |
| Logging interval | 20 steps | 20 steps |
| Data loading workers | 2 | 2 |
| Dataset preprocessing processes | 2 | 2 |

**Table 4.3:** LoRA fine-tuning hyperparameters for Polyp VQA

The visual-grounded setting uses a longer maximum sequence length of 1024 because each example contains two visual inputs: the original endoscopic image and the segmentation-derived grounding image. In contrast, the original-image LoRA setting uses a maximum length of 512 because it only includes one image and the question-answer text.

The number of epochs is also different between the two settings. Original-image LoRA is trained for one epoch on the full training set, while the visual-grounded setting is trained for three epochs on the filtered subset. Since the mask-filtered subset is substantially smaller than the full VQA training set, the additional epochs are used to keep the number of training iterations more comparable between the two training strategies. This reduces optimization-step count as a potential confounding variable when comparing original-image fine-tuning and visual-grounded fine-tuning. Training and validation loss curves did not show signs of overfitting across the visual-grounded epochs, supporting the use of additional epochs on the smaller subset.

The original-image setting trains each model to answer from the endoscopic frame and question only. The visual-grounded setting extends the input with segmentation-derived evidence, such as a mask or overlay. Fair comparison requires controlling the evaluation split and LoRA strategy so that two effects can be separated: the difference between base-model backbones and the effect of explicit segmentation guidance.

The output format is kept consistent across training and evaluation. Short factual answers are easier to evaluate with lexical metrics, while longer explanatory answers may better reflect clinical reasoning but are harder to score with BLEU or ROUGE [47, 48]. BLEU rewards $n$-gram precision, and ROUGE-L captures longest-common-subsequence overlap; both are useful, but neither can fully decide whether a medical answer is clinically appropriate. For this reason, LLM Judge is included as a complementary evaluation method [49].

### 4.2.2 Evaluation Protocol

The evaluation protocol is kept as a compact bridge between the datasets and the results. It is necessary because each module has a different output type: text reasoning is evaluated by relation accuracy and multi-turn awareness, segmentation by mask overlap, Polyp VQA by lexical and judge-based answer quality, and memory by continuity-oriented metrics. Table 4.4 summarizes the evaluated components, comparison settings, and primary metrics.

| Component | Evaluation Data | Compared Settings | Metrics |
|---|---|---|---|
| **Text Module** | One-hop relation and multi-turn awareness benchmarks | KG + MedLLM; RAG + MedLLM; MedLLM | One-Hop Accuracy; Multi-Turn Awareness Score |
| **Polyp Segmentation** | BKAI-IGH NeoPolyp-Small | Segmentation models with and without post-processing | Dice Score [46] |
| **Polyp VQA** | Kvasir-VQA-X1 and its mask-matched subset | Base VLMs; LoRA fine-tuning; visual-grounded LoRA | BLEU-4; ROUGE-L; LLM-Judge Score [47, 48, 49] |
| **Memory Layer** | 30 Vietnamese multi-session scenarios | M0 no memory; M1 seeded memory; M2 extracted memory | Extraction P/R; routing accuracy; repeated-question avoidance; answer quality |

**Table 4.4:** Evaluation protocol for the main system components

For Polyp VQA, full-dataset and subset-based comparisons are separated. Original-image VQA results on the full Kvasir-VQA-X1 evaluation set are not treated as directly comparable to visual-grounded results unless both are evaluated on the same mask-matched subset. This avoids attributing differences caused by data filtering to the grounding method itself. Numerical metrics are interpreted together with qualitative behavior because medical AI outputs cannot be judged reliably by a single score.

## 4.3 Results and Analysis

### 4.3.1 Text Module Results and Analysis

The Text Module is evaluated using one-hop relation questions and multi-turn awareness cases. The one-hop setting measures whether the system can answer relation-oriented medical questions when the relevant relation is available, while the multi-turn setting measures whether the system recognizes that a patient-specific query is underspecified and should trigger follow-up clarification. These two metrics therefore capture different behaviors and should not be averaged into a single score.

Three settings are compared. **MedLLM** represents the base language-model setting without external KG or RAG evidence. **RAG + MedLLM** adds retrieved unstructured medical documents before answer generation. **KG + MedLLM** adds structured medical relations from the Knowledge Graph before MedGemma CoT assessment and synthesis. Table 4.5 summarizes the comparison. Scores are reported as percentages, and higher values indicate better performance.

| Model Variant | One-Hop Accuracy | Multi-Turn Awareness Score |
|---|---|---|
| **KG + MedLLM** | **86.95%** | **70.83%** |
| RAG + MedLLM | 67.97% | 40.38% |
| MedLLM | 60.87% | 50.72% |

**Table 4.5:** Text Module benchmark results

The KG + MedLLM setting achieves the best performance in both One-Hop Accuracy and Multi-Turn Awareness Score. Compared with the base MedLLM setting, KG + MedLLM improves One-Hop Accuracy from 60.87% to 86.95%, a gain of +26.08 percentage points. It also improves Multi-Turn Awareness Score from 50.72% to 70.83%, a gain of +20.11 percentage points. Compared with RAG + MedLLM, the KG-based setting is higher by +18.98 percentage points on One-Hop Accuracy and +30.45 percentage points on Multi-Turn Awareness. These differences suggest that structured medical relations help the model answer relation-oriented questions more accurately and also support better recognition of insufficient patient context.

RAG + MedLLM performs better than the base model on one-hop reasoning, increasing One-Hop Accuracy from 60.87% to 67.97% (+7.10 percentage points), but it is weaker on Multi-Turn Awareness, decreasing from 50.72% to 40.38% (-10.34 percentage points). This contrast is important. Retrieved text can provide useful explanatory evidence, but it may also encourage the model to produce a direct answer when the original clinical question remains incomplete. The base MedLLM setting lacks external grounding and therefore performs worse on structured relation reasoning, but it is not affected by potentially over-confident retrieved context in the same way.

This separation matters for safety. A model can achieve reasonable One-Hop Accuracy by answering relation-oriented questions, but still behave poorly in a consultation setting if it answers too early when the patient context is incomplete. A medical assistant should therefore perform well on both dimensions: it should answer when the evidence and clinical context are sufficient, and it should ask targeted follow-up questions when important variables are missing.

The stronger performance of KG + MedLLM suggests that explicit relations help the model identify clinically relevant links more reliably than text retrieval alone in this benchmark. However, the result should not be interpreted as showing that KG is always superior to RAG. The benchmark emphasizes relation-oriented questions and insufficient-context awareness, which naturally favors structured evidence. RAG remains useful for explanatory responses, disease overviews, treatment descriptions, and broader medical education. The practical conclusion is that the two sources should be coordinated rather than treated as mutually exclusive.

The lower Multi-Turn Awareness Score of RAG + MedLLM supports the need for an explicit sufficiency check. Retrieved evidence can make the model appear grounded while the original question remains underspecified. The MedGemma CoT information-sufficiency step is therefore necessary after KG/RAG context selection and before final answer synthesis.

The Multi-Turn Awareness Score should not be read as direct evidence that the system has solved full path-level KG multi-hop reasoning. The benchmark mainly measures whether the system recognizes insufficient context and avoids a premature answer. The advantage of KG-based multi-hop reasoning is therefore discussed as an architectural capability: the graph can support structured chains such as symptom $\rightarrow$ disease $\rightarrow$ treatment, while the evaluated multi-turn behavior measures whether the system knows when more user context is required before using those relations in a patient-facing response.

The base MedLLM result in Table 4.5 shows that a language model alone can answer some medical questions but performs worse when structured external knowledge is required. Therefore, the benchmark motivates placing the LLM inside a retrieval and graph-augmented pipeline rather than using it as the only reasoning source.

**Multi-turn clarification case analysis.** The difference between a traditional single-turn response and the proposed multi-turn clarification behavior is illustrated in Figures 4.5 and 4.6. These two examples use the same general symptom scenario, but the response strategies are different. They should be interpreted as qualitative case analysis that explains the failure mode and desired behavior behind the Multi-Turn Awareness Score, rather than as additional quantitative benchmark results.

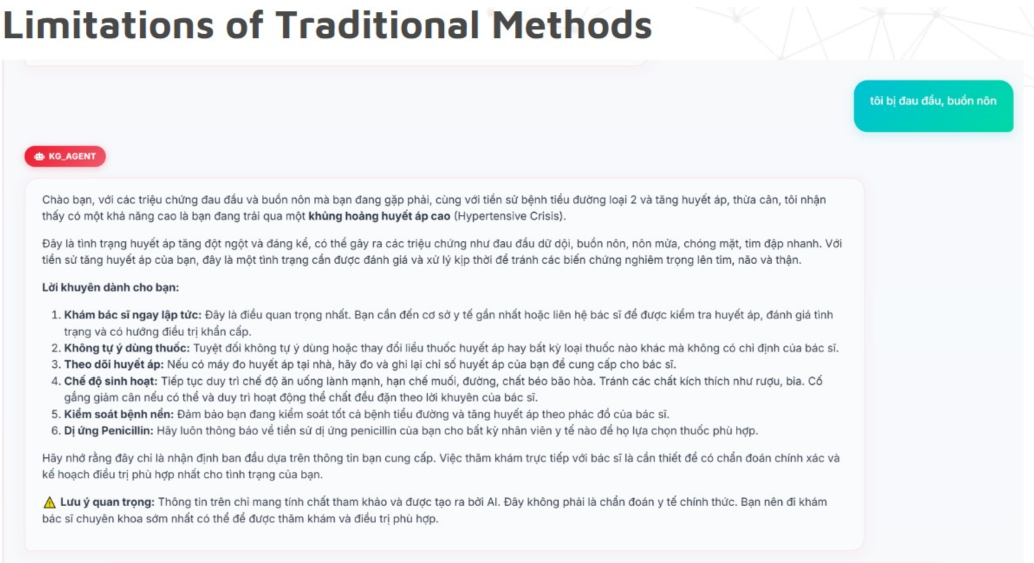


**Figure 4.5:** Traditional single-turn approach with limited clinical context

Figure 4.5 shows the main weakness of a traditional single-turn medical assistant. The user provides a short symptom description, such as headache and nausea, but the system immediately produces a relatively strong medical interpretation. Even if the response includes disclaimers, the reasoning is still based on an incomplete clinical picture. Important variables such as current blood pressure, onset time, severity, neurological symptoms, medication adherence, and warning signs are not yet available. As a result, the model may overcommit to one hypothesis or provide advice that is not sufficiently adapted to the actual situation.

This failure mode is particularly important in medical AI because symptoms are rarely specific by themselves. Headache and nausea can be associated with many conditions, ranging from mild fatigue or migraine to hypertensive urgency, infection, medication side effects, or neurological emergencies. A single-turn answer may sound helpful because it is detailed, but detail does not guarantee clinical appropriateness. The core problem is that the model is answering before it has enough discriminative information.

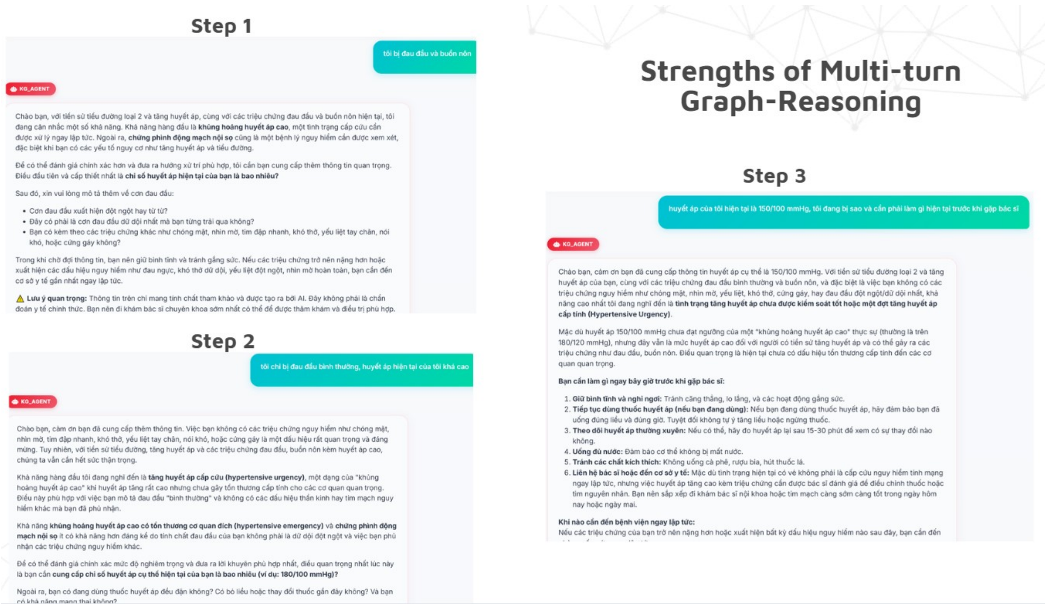


**Figure 4.6:** Multi-turn clarification with iterative context collection and safer response refinement

Figure 4.6 demonstrates the advantage of the multi-turn clarification workflow. Instead of producing a final answer immediately, the system first identifies missing clinical information and asks a focused follow-up question. When the user provides additional context, such as a blood pressure value or the absence of severe neurological symptoms, the system updates its interpretation. The answer then becomes more specific: it can distinguish between hypertensive emergency, hypertensive urgency, and a high but not immediately life-threatening blood pressure state.

This iterative process improves the response in three ways. First, it improves **clinical specificity** because the system can condition its answer on newly provided variables. Second, it improves **safety** because red flags are checked before giving ordinary advice. Third, it improves **explainability** because each follow-up question corresponds to a missing piece of information that affects the final decision. In this sense, multi-turn awareness is not only a conversational feature; it is a safety mechanism for controlling when the system is allowed to answer.

The comparison also clarifies the role of the Conversation Agent. Its purpose is not to ask many questions for their own sake, but to ask the next question that most reduces clinical uncertainty. A good follow-up question should be short, relevant, and oriented toward high-impact variables such as red flags, vital signs, duration, severity, medication use, and patient history. Once sufficient information is available, the system can move from clarification to evidence retrieval and MedGemma CoT synthesis.

### 4.3.2 Polyp Segmentation Results and Analysis

Table 4.6 presents Dice Score results for several segmentation models on the BKAI-IGH NeoPolyp-Small segmentation task, with and without the proposed post-processing step. Dice Score is used here as an overlap-based metric between the predicted mask and the ground-truth annotation, so a higher value indicates better agreement in pixel-level lesion localization and class-aware mask prediction.

| Models | DICE Score | |
|---|---|---|
| | Without Processing | With Processing |
| PraNet | 80.02% | 82.94% |
| DeepLabV3+ (RegNetx320) | **82.56%** | **86.33%** |
| DeepLabV3+ (ResNet50) | 81.12% | 85.62% |
| EMCAD (PVT2-b2) | 80.93% | 83.76% |

**Table 4.6:** Dice Score comparison for polyp segmentation models

Post-processing improves segmentation consistency across all models for which post-processed results are reported. PraNet increases from 80.02% to 82.94%, DeepLabV3+ (RegNetx320) increases from 82.56% to 86.33%, DeepLabV3+ (ResNet50) increases from 81.12% to 85.62%, and EMCAD (PVT2-b2) increases from 80.93% to 83.76%. These correspond to gains of +2.92, +3.77, +4.50, and +2.83 percentage points, respectively. The average improvement over these four post-processed models is therefore approximately +3.51 percentage points. U-Net and U-Net++ are included as baseline segmentation results, but no post-processing value is reported for these two models, so they are not included in the gain analysis.

Among the compared settings, DeepLabV3+ (RegNetx320) achieves the strongest final result with 86.33% Dice Score after post-processing. DeepLabV3+ (ResNet50) obtains the largest absolute improvement, increasing by +4.50 percentage points. This pattern suggests that the refinement step is useful not only for the best-performing backbone, but also for models whose raw predictions contain more class inconsistency inside the detected lesion region.

The improvement is consistent with the post-processing method described in the Vision Reasoning Module. Connected component analysis changes the interpretation of the mask from isolated pixel predictions to region-level candidates, and region-wise color dominance refinement reduces mixed red/green class fragments inside a connected lesion. This is especially relevant in the BKAI-IGH NeoPolyp setting because a pixel can be spatially inside the lesion but still counted as wrong if its neoplastic/non-neoplastic label is incorrect.

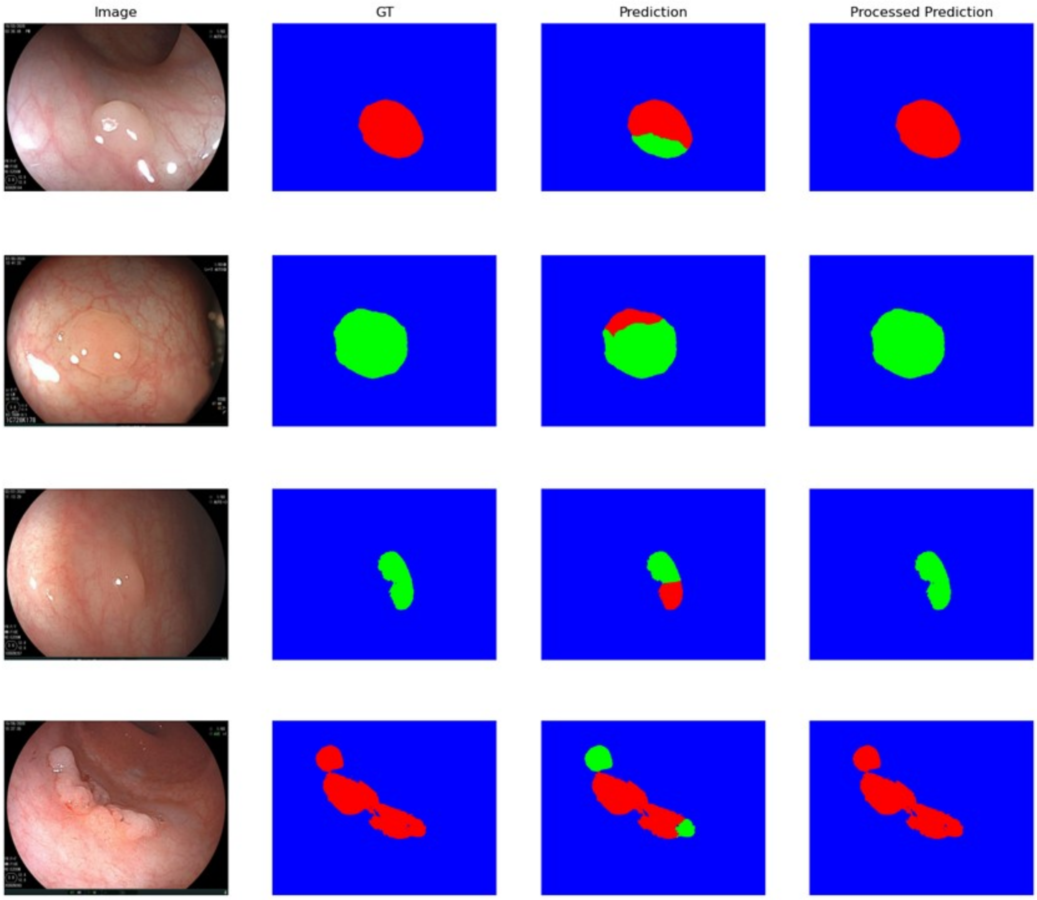


Figure 5: Comparision Result of The Masking Segmentation with The Technique

**Figure 4.7:** Visualization of raw and post-processed polyp segmentation masks

The qualitative examples in Figure 4.7 align with the quantitative trend in Table 4.6. Each row compares the original colonoscopy image, the ground-truth mask, the raw prediction, and the processed prediction. In the first two examples, the raw prediction locates the lesion but assigns a small part of the connected region to the wrong class. After refinement, the minority-class artifact is removed and the processed mask becomes closer to the ground truth. In the third and fourth examples, the same effect appears in more irregular connected regions: the method preserves the approximate lesion geometry while making the class label inside the region more consistent.

The largest gains therefore appear when the raw prediction is close to the correct region but contains minority-class artifacts. In these cases, $C_k(R_i)$ identifies the dominant class, and the update rule for $M'(x, y)$ improves overlap by regularizing the label inside the connected component. This explains why the method can improve Dice Score without changing the underlying segmentation backbone: it corrects local class fragmentation after the model has already detected the lesion region.

The method should therefore be interpreted as a consistency refinement module, not as a replacement for the segmentation model. It cannot recover a missed lesion, and it may incorrectly merge touching regions that actually belong to different classes. For downstream Polyp VQA, its main benefit is a clearer overlay for visual grounding and human review; its main risk is that an incorrect refined mask may guide the VQA model toward wrong evidence. This motivates the separate evaluation of original-image VQA and segmentation-grounded VQA.

### 4.3.3 Polyp VQA Fine-tuning Results and Analysis on Kvasir-VQA-X1

The Polyp VQA VLM is evaluated using BLEU-4, ROUGE-L, and an LLM-as-a-judge protocol [47, 48, 49]. BLEU-4 and ROUGE-L measure lexical overlap with the reference answer, while the judge-based score is used to assess semantic and clinical answer quality. The judge model is `google/medgemma-27b-it`. For each prediction, the judge receives the question, ground-truth answer, and model prediction, and returns four scores from 1 to 5: factual accuracy, completeness, clinical relevance, and no hallucination. The weighted LLM-Judge Score is computed as:

$$\begin{aligned}\text{LLM-Judge Score} = {} & 0.40 \times \text{Factual Accuracy} + 0.40 \times \text{Completeness} \\ & + 0.10 \times \text{Clinical Relevance} + 0.10 \times \text{No Hallucination}.\end{aligned}$$

This weighting gives the highest importance to factual correctness and coverage of the reference answer, while still accounting for clinical language quality and hallucination control. This is appropriate for medical VQA because an answer that is fluent but factually wrong or incomplete should not be considered reliable.

| Model Setting | Samples | BLEU-4 | ROUGE-L | LLM-Judge Score |
|---|---|---|---|---|
| MedGemma-4B-it base | 1000 | 0.0055 | 0.1154 | 2.7521 |
| MedGemma-4B-it + LoRA | 1000 | 0.3657 | 0.6936 | 4.6351 |
| Qwen3.5-4B base | 1000 | 0.0261 | 0.2078 | 3.4720 |
| Qwen3.5-4B + LoRA | 1000 | 0.1935 | 0.5211 | 4.5341 |
| MedGemma-4B-it + LoRA only | 305 | – | – | 4.4269 |
| MedGemma-4B-it + LoRA + visual grounding | 305 | – | – | 4.5751 |

**Table 4.7:** Polyp VQA evaluation on Kvasir-VQA-X1

The clearest pattern in Table 4.7 is that LoRA fine-tuning is the main driver of performance improvement. MedGemma-4B-it improves from a BLEU-4 score of 0.0055 and an LLM-Judge Score of 2.7521 to 0.3657 and 4.6351, respectively. Qwen3.5-4B also benefits strongly from LoRA, increasing from a BLEU-4 score of 0.0261 and an LLM-Judge Score of 3.4720 to 0.1935 and 4.5341. In practical terms, the base VLMs can process the image-question input, but they are not yet aligned with the answer style and clinical content distribution of Kvasir-VQA-X1. Lightweight task adaptation substantially improves both lexical matching and semantic answer quality.

Between the two backbones, MedGemma-4B-it achieves stronger fine-tuned performance than Qwen3.5-4B. This is consistent with the expectation that a medically oriented VLM backbone is better aligned with endoscopic visual concepts and clinical answer generation. Qwen3.5-4B remains a strong baseline after fine-tuning, but its lower BLEU-4 and ROUGE-L suggest that it may be less aligned with the dataset's medical visual-language distribution.

The visual grounding comparison must be interpreted on the mask-matched subset only. As shown in Table 4.2, only 1,336 of the 6,212 training images have usable segmentation masks, reducing the training samples from 143,594 to 42,382 for the grounded setting. Therefore, the fair comparison is not between the 1000-sample MedGemma LoRA result and the visual-grounded result.

Instead, the correct baseline is the MedGemma LoRA-only model evaluated again on the same 305 evaluation samples that have matched visual grounding inputs. Under this controlled comparison, visual grounding improves the LLM-Judge Score from 4.4269 to 4.5751. The gain is modest, but it is meaningful because the grounded model is being tested against a strong LoRA-only baseline on the same subset. The result supports the idea that segmentation masks can provide useful spatial evidence for VQA when the mask is available and aligned with the question.

The original-image setting evaluates whether the VLM can identify relevant visual evidence without additional guidance. The visual-grounded setting evaluates whether a segmentation mask improves answer quality by focusing attention on the polyp region. Since only a subset of samples contains matched segmentation masks, the visual grounding analysis is conducted on 305 mask-matched samples.

| Setting | Factual Acc. | Completeness | Clinical Rel. | No Halluc. | Overall |
|---|---|---|---|---|---|
| LoRA only | 4.2918 | 4.3607 | 4.8984 | 4.7607 | 4.4269 |
| Visual-grounded LoRA | 4.4951 | 4.5016 | 4.9246 | 4.8393 | 4.5751 |
| Difference | +0.2033 | +0.1409 | +0.0262 | +0.0786 | +0.1482 |

**Table 4.8:** LLM Judge comparison between LoRA-only and visual-grounded MedGemma on the mask-matched subset

The criterion-level judge results in Table 4.8 show improvement across all four judge criteria. The largest gain appears in factual accuracy (+0.2033), followed by completeness (+0.1409). This pattern is important because it suggests that the mask is not merely changing the wording of the answer. It appears to help the model select more correct visual evidence and include more of the expected clinical content. The smaller gain in clinical relevance (+0.0262) is also reasonable because the LoRA-only model already has a very high clinical relevance score of 4.8984, leaving little room for improvement.

The hallucination score also improves from 4.7607 to 4.8393. This suggests that explicit spatial grounding may reduce the tendency to add unsupported visual claims. In medical VQA, this is valuable because a confident but visually unsupported answer can be unsafe even when it uses correct clinical terminology.

The controlled comparison in Table 4.7, together with the criterion-level breakdown in Table 4.8, supports the design choice of connecting the Segmentation Agent with the Polyp VQA Agent. The segmentation mask acts as an intermediate visual evidence signal, allowing the VQA model to focus on the lesion area instead of relying only on global image context or dataset priors. However, the result should not be interpreted as proof that visual grounding is always beneficial. The grounded setting depends on mask availability and mask quality. If the segmentation model misses the lesion, highlights irrelevant tissue, or assigns an incorrect region, the same grounding mechanism may guide the VQA model toward the wrong evidence. For this reason, visual grounding should be treated as a useful but conditional intervention, and future evaluation should include failure cases where incorrect masks degrade the answer.

### 4.3.4 Memory Layer Results and Analysis

The Memory Layer is evaluated on 30 Vietnamese multi-session medical scenarios. The experiment compares three settings: M0 disables long-term memory, M1 uses seeded ground-truth memories as an oracle-style upper bound for retrieval, and M2 stores memories extracted automatically from prior conversation sessions. This design separates three effects: the value of having no memory, the potential value of perfectly prepared memory, and the practical performance of the implemented memory extraction pipeline.

| Setting | Cases | Extraction P/R | Routing Acc. | Repeated-Q Avoid. | Answer Quality |
|---|---|---|---|---|---|
| M0: no memory | 30 | – | 86.67% | – | 3.27 |
| M1: seeded memory | 30 | 1.00 / 1.00 | 96.67% | 100.00% | 4.47 |
| M2: extracted memory | 30 | 0.58 / 0.80 | 96.67% | 83.33% | 4.27 |

**Table 4.9:** Memory Layer evaluation results on 30 multi-session scenarios

Table 4.9 shows that long-term memory improves the system mainly through contextual continuity rather than through large changes in routing. Without memory, M0 achieves a routing accuracy of 86.67%, but it does not have access to patient-specific facts from previous sessions. When memory is available, both M1 and M2 reach 96.67% routing accuracy, corresponding to a 10 percentage-point improvement over M0. This indicates that the Decision Agent already performs well for many queries without memory, while remembered patient context helps in cases where the current request is underspecified or depends on prior information.

The strongest improvement appears in answer quality. The average answer quality increases from 3.27 in M0 to 4.47 in M1 and 4.27 in M2. The reason is easy to see in consultation-style dialogue: without memory, the assistant often has to ask generic background questions again; with memory, it can reuse known allergies, chronic conditions, medication use, or prior symptoms and move to a more targeted next question. This is also reflected in repeated-question avoidance: M1 reaches 100.00%, while M2 reaches 83.33%. The result supports the thesis argument that memory is useful as a continuity mechanism, especially when the same patient context remains relevant across sessions.

The difference between M1 and M2 is important for interpreting the practical maturity of the module. M1 uses seeded reference memories, so its extraction precision and recall are both 1.00. This provides a controlled setting in which MedGemma CoT and the routed task agent receive the expected patient context. M2, which relies on automatic extraction from conversation, obtains 0.58 precision and 0.80 recall. This means that the automatic memory pipeline captures most expected durable facts, but it also stores or retrieves additional facts that are not part of the reference memory set. The lower answer quality of M2 compared with M1 shows that imperfect memory extraction can affect downstream answer synthesis, even when routing accuracy remains unchanged.

Latency is the main cost of adding memory. Retrieval itself contributes about 1.26-1.51 seconds on average, while routing latency increases from 3.12 seconds to approximately 4.65 seconds because the Decision Agent receives a longer memory-augmented context. The remaining latency increase comes from full-system answer synthesis under richer context. Thus, the memory layer improves response quality and continuity, but it also introduces a clear efficiency trade-off.

Across difficulty levels, M1 and M2 achieve perfect routing accuracy on medium cases, while hard cases remain at 83.33% for all three settings. This pattern suggests that memory helps resolve ordinary contextual incompleteness, but it does not fully solve difficult routing cases involving adversarial prompts, ambiguous references, or safety-sensitive judgment. Therefore, memory should be interpreted as supporting evidence for routing and answer synthesis, not as a substitute for stronger safety policies, contradiction handling, and medical uncertainty assessment.

Overall, the memory evaluation supports the use of patient-scoped long-term memory in the proposed multi-agent architecture. Memory improves personalization, reduces repeated questioning, and increases judged answer quality. At the same time, the gap between M1 and M2 shows that automatic memory extraction still requires refinement, especially in precision, relevance filtering, recency handling, and deletion or correction mechanisms. These findings are consistent with the design principle used throughout the thesis: memory should provide contextual support, but it should not be treated as verified clinical truth or allowed to override current evidence.

# CHAPTER 5. DISCUSSION AND CONCLUSION

## 5.1 Overall Discussion

The main strength of the system is its modular AI architecture. Instead of relying on a single model, the system combines specialized components for routing, text reasoning, graph reasoning, retrieval, visual analysis, VQA, memory, and validation. This design improves extensibility and makes the reasoning process easier to inspect because each intermediate decision can be associated with a specific module.

The architecture also supports complementary reasoning sources. KG provides structured medical relations and can organize evidence paths such as symptom $\rightarrow$ disease $\rightarrow$ treatment, while RAG provides explanatory text for patient-facing answers. Web Search provides fallback coverage when internal evidence is weak, and memory provides patient-specific continuity. In the Vision Module, segmentation provides localized evidence, post-processing improves mask consistency, and the VQA agent converts original-image and mask-guided visual input into an answer. This combination allows the system to handle a wider range of medical-assistant tasks than a single text-only chatbot.

Another strength is that the system makes several uncertainty points explicit. MedGemma CoT-based sufficiency checks identify incomplete symptom descriptions after KG/RAG agents have selected relevant context, and the system can then decide whether to synthesize an answer, ask a follow-up question, or use fallback search. This behavior is evaluated through Multi-Turn Awareness Score rather than treated as direct proof of full KG path reasoning. Routing determines whether a specialized agent is appropriate. Visual grounding is evaluated on a mask-matched subset rather than assumed to be universally beneficial. Memory is evaluated through no-memory, seeded-memory, and extracted-memory settings, which separates the value of continuity from the current limitations of automatic memory extraction. These design choices improve transparency and make the system easier to evaluate.

## 5.2 Limitations

### 5.2.1 Text Module

The Text Module depends on the coverage and quality of its data sources. RAG may retrieve irrelevant documents, and the Knowledge Graph may not contain all relevant medical relations. Multi-turn clarification also depends on correctly identifying missing information. The current text benchmark evaluates one-hop relation answering and multi-turn awareness, but it does not yet evaluate complete path-level KG reasoning across chains such as symptom $\rightarrow$ disease $\rightarrow$ treatment $\rightarrow$ warning sign.

Vietnamese medical language adds additional challenges. Users may describe symptoms in informal language, use regional terms, or mix Vietnamese with English medical terminology. If query expansion does not capture these variations, retrieval quality may decrease. If expansion is too broad, the system may retrieve clinically unrelated contexts. This trade-off affects both answer accuracy and safety.

The Text Module also cannot guarantee clinical correctness. Even when KG and RAG provide useful evidence, MedGemma CoT still performs the final synthesis with an LLM and may omit caveats, understate uncertainty, or over-generalize from incomplete context. Human validation and expert review remain necessary for high-risk cases.

### 5.2.2 Vision Module

The Vision Module is limited by the quality of segmentation, mask availability, and VQA fine-tuning data. General medical image interpretation is broad and difficult to validate across all modalities. Polyp VQA is more focused, but the experimental result is tied to Kvasir-VQA-X1 and to the subset of images that have usable segmentation masks.

The segmentation results show that region-level post-processing improves Dice Score, with DeepLabV3+ (RegNetx320) increasing from 82.56% to 86.33%. This supports the usefulness of connected-component and color-dominance refinement for making masks more consistent. However, the method regularizes an existing prediction rather than detecting missed lesions. It may improve a noisy but mostly correct region, but it cannot recover a lesion that the base model fails to segment.

Segmentation errors can propagate to downstream VQA. If the mask misses the polyp region, highlights irrelevant tissue, or assigns the wrong class, the grounded VQA input may mislead the model. Therefore, visual grounding is not automatically safer than original-image VQA. The results section shows a positive controlled effect on the mask-matched subset, where MedGemma LoRA with visual grounding improves the LLM-Judge Score from 4.4269 to 4.5751 and improves factual accuracy, completeness, clinical relevance, and no-hallucination criteria. This should be interpreted as conditional evidence that masks help when they are available and aligned with the question, not as proof that every mask-guided answer is safer.

The General Diagnosis Agent has broader coverage but weaker guarantees. Without modality-specific fine-tuning and evaluation, its outputs should remain descriptive and cautious. This agent is useful for system extensibility, but the thesis does not claim comprehensive diagnostic performance across all medical image types.

### 5.2.3 Kvasir-VQA-X1 and Domain Transfer

Kvasir-VQA-X1 is useful for endoscopic VQA, but dataset coverage and annotation style may not fully match real deployment images or user questions. Metrics such as BLEU and ROUGE-L may not capture clinical correctness, so LLM Judge and qualitative analysis are also needed. In the current results, LoRA fine-tuning is the main driver of VQA improvement: MedGemma-4B-it increases from 2.7521 to 4.6351 in LLM-Judge Score, and Qwen3.5-4B increases from 3.4720 to 4.5341. This means that the conclusion should emphasize both task adaptation and visual grounding, not visual grounding alone.

Domain transfer is a central concern. A model fine-tuned on Kvasir-VQA-X1 may learn the style and distribution of that dataset but still struggle with images from different devices, lighting conditions, bowel preparation quality, or clinical populations. User questions in deployment may also be longer, less structured, and more ambiguous than dataset questions. These differences should be considered when interpreting evaluation results.

The two VQA settings also introduce different risks. The original-image setting depends on the VLM's own attention and visual reasoning. The grounded setting depends on both segmentation quality and the VLM's ability to use the overlay correctly. A fair discussion must therefore analyze not only which setting scores higher but also why it succeeds or fails. Because visual grounding is evaluated on 305 mask-matched evaluation samples, it should not be directly compared with full-set LoRA results on 1000 samples. The fair comparison is the LoRA-only versus visual-grounded setting on the same mask-matched subset.

### 5.2.4 Memory Layer

Memory can be stale, incomplete, or incorrectly extracted. Patient-reported facts may change over time. Therefore, memory should be used as supporting context only and should not override current user input or medical evidence. The memory evaluation shows that long-term memory improves answer quality from 3.27 in the no-memory setting to 4.47 with seeded memory and 4.27 with extracted memory. It also improves routing accuracy from 86.67% to 96.67% when memory is available. These results support memory as a useful continuity mechanism, but not as a substitute for current evidence or expert judgment.

A memory layer can also introduce privacy and governance concerns. Even clinically useful facts may be sensitive. The system should avoid storing unnecessary personally identifiable information and should support correction or deletion of stored facts. In the current thesis scope, memory is evaluated as a system integration feature rather than as a complete clinical record management system. The gap between seeded memory and extracted memory is especially important: extracted memory reaches 0.58 precision and 0.80 recall, which means it captures many useful facts but also stores or retrieves extra information that can affect answer synthesis.

Another limitation is that memory retrieval may surface irrelevant old facts. If MedGemma CoT or another routed task agent uses such facts without checking relevance, personalization can become distracting or unsafe. The agent should only use memory when it is directly related to the current question, and the retrieved memory should remain subordinate to the user's current statement and external medical evidence. Memory also introduces latency: retrieval and longer memory-augmented prompts increase routing and full-system response time. This is an acceptable trade-off for continuity in many consultation settings, but it should be optimized before deployment.

### 5.2.5 Evaluation and Safety

The evaluation is component-oriented and system-oriented, but it is not a clinical trial. The system should be positioned as medical decision support, not as a replacement for physicians. More rigorous expert evaluation is necessary before any clinical use.

The metrics used in this thesis capture important but incomplete aspects of performance. Dice Score measures segmentation overlap but does not fully measure clinical usefulness or downstream VQA reliability. BLEU and ROUGE-L measure text similarity but not necessarily medical correctness. LLM Judge can evaluate semantic quality but may contain evaluator bias, even when a medical judge model is used. Memory metrics such as extraction precision/recall, repeated-question avoidance, and answer quality capture continuity behavior but do not prove clinical safety. One-Hop Accuracy and Multi-Turn Awareness Score measure relation-oriented answering and insufficient-context awareness, but they cannot cover all clinical scenarios or prove full path-level KG reasoning. Multi-Turn Awareness should therefore be interpreted as a safety behavior: the model recognizes when a patient-specific answer requires more context, rather than simply producing a longer conversation.

For these reasons, the results should be interpreted as evidence of technical feasibility and comparative system behavior, not as proof of clinical readiness. Future evaluation should include medical expert review, larger datasets, prospective testing, and safety analysis under realistic user interactions.

## 5.3 Conclusion

### 5.3.1 Summary of Achievements

This thesis demonstrates that a medical assistant can be made more controllable when routing, evidence retrieval, visual grounding, memory retrieval, and validation are separated into explicit workflow components. MedGemma CoT is used for text-side sufficiency assessment and answer synthesis rather than as an isolated answer generator. The system improves observability and safety behavior by allowing intermediate evidence to be inspected, distinguishing evidence availability from clinical sufficiency, preserving patient context without treating it as verified diagnosis, and evaluating visual grounding as a conditional intervention.

Experimentally, the thesis shows five main outcomes. First, the Text Module benchmark shows that KG + MedLLM achieves the strongest One-Hop Accuracy and Multi-Turn Awareness Score, supporting the usefulness of structured evidence and explicit insufficient-context detection. This should be interpreted as evidence for relation-oriented answering and multi-turn awareness, while full KG path-level reasoning remains a future evaluation target. Second, connected-component and color-dominance post-processing improves segmentation consistency for colonoscopy polyp masks, with the strongest DeepLabV3+ setting reaching 86.33% Dice Score after post-processing. Third, LoRA fine-tuning substantially improves Polyp VQA performance for both MedGemma-4B-it and Qwen3.5-4B, showing that task adaptation is essential before using compact VLMs for endoscopic VQA. Fourth, segmentation-grounded VQA improves judge-based answer quality on the mask-matched subset, especially in factual accuracy and completeness, but this gain is conditional on mask availability and mask quality. Fifth, patient-scoped memory improves personalization, routing accuracy, repeated-question avoidance, and answer quality, while adding latency and relevance-filtering trade-offs. Together, these results support the thesis argument that modular grounding and evidence control can improve medical-assistant behavior at the system level.

### 5.3.2 Technical Contributions

The technical contributions are:

- a graph-orchestrated multi-agent architecture that separates routing, text reasoning, vision reasoning, memory retrieval, and validation;
- a Text Module that combines multi-turn clarification, KG/RAG evidence construction, MedGemma CoT sufficiency assessment and answer synthesis, and fallback search;
- a Vision Module that links three-class polyp segmentation, mask post-processing, original-image Polyp VQA, and segmentation-grounded Polyp VQA;
- a region-level post-processing method based on connected components and color dominance for improving multi-class polyp mask consistency;
- LoRA fine-tuned Polyp VQA models based on MedGemma-4B-it and Qwen3.5-4B, evaluated under original-image and mask-matched visual-grounded settings;
- a patient-scoped memory layer and human-in-the-loop validation mechanism for contextual continuity, feedback, repeated-question reduction, and review.

## 5.4 Future Work

Future work should focus on expanding Vietnamese medical datasets, improving Knowledge Graph coverage, strengthening RAG retrieval and reranking, improving visual grounding robustness, refining LoRA fine-tuning for VQA, adding expert validation, and developing stronger memory governance mechanisms. For the Text Module, an important next step is to build a dedicated path-level KG benchmark for chains such as symptom $\rightarrow$ disease $\rightarrow$ treatment and symptom $\rightarrow$ disease $\rightarrow$ warning sign, separately from Multi-Turn Awareness Score. Multi-turn evaluation should also cover longer clinical dialogues that test relevant follow-up questions, stopping behavior when enough context is available, and safety boundaries during answer synthesis. For the Vision Module, future work should evaluate grounding under segmentation failure cases, expand mask-matched data coverage, and test whether improved masks consistently improve VQA behavior rather than only Dice Score. For the Memory Layer, future work should improve extraction precision, recency handling, contradiction resolution, user correction, deletion workflows, and latency control. Clinical evaluation with medical professionals is also necessary before real-world deployment.

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